

Optimizing Lettuce Growth in Controlled High-Humidity Environments

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Abstract

Successful cultivation of lettuce (*Lactuca sativa*) in indoor agriculture relies on precise control of environmental factors to optimize crop yields. Vertical farms often face challenges with humidity management, as net transpiration rates from high-density lettuce cultivation can exceed the capacity of dehumidification and cooling systems. The result are elevated humidity conditions that promote physiological disorders of lettuce, particularly necrosis of the outer tips of younger leaves known as “tipburn”. This thesis observed the effects of different lighting treatments and air velocities on lettuce at varying levels of humidity as a means of mitigating deleterious effects of elevated humidity. Romaine lettuce (*Lactuca sativa*, cv. Teton) was hydroponically cultivated in three growth chambers under three constant relative humidities (RH): 70%, 80%, and 90%. Within each chamber, three lighting treatments were tested: white light (WL), white light supplemented with far red (WFR), and wide amber supplemented with far red (WA). Each growth tray included an air movement system constructed using a polyethylene tube attached to a duct fan to deliver two air velocity zones: low (0.5 m s^{-1}) and high (1.0 m s^{-1}). Under the air movement system, average air velocity experienced by plants ranged from $0.23\text{--}3.0 \text{ m s}^{-1}$. Tipburn reduction was highly effective even at the highest RH of 90%, as air movement ensured transpiration remained highly functional while preventing physiological disorders linked to diminished stomatal response under prolonged high humidity conditions. Results indicate that far-red supplementation to amber wavelengths may steer plant development towards leaf expansion, especially under elevated humidities. Far-red likely improved CO_2 assimilation under a low Red: Blue ratio (0.15), with enhancement of stomatal conductance and comparable net assimilation to WA and WL supporting prior research on the promotive effects of blue light on stomatal regulation as well as potential synergistic interactions with far-red. The results of this study indicate that air movement systems providing air velocities from $0.5\text{--}1.0 \text{ m s}^{-1}$ at tray level are effective in eliminating tipburn in lettuce at an RH up to 90% without triggering additional stress responses. The benefits of air movement provide growers flexibility to modify lighting spectra to further optimize plant growth without concern for the negative effects of elevated humidity.

Résumé

La réussite de la culture de laitues (*Lactuca sativa*) en agriculture d'intérieur repose sur un contrôle précis des facteurs environnementaux afin d'optimiser les rendements des cultures. Les fermes verticales sont souvent confrontées à des problèmes de gestion de l'humidité, car les taux de transpiration nette des cultures de laitues à haute densité peuvent dépasser la capacité des systèmes de déshumidification et de refroidissement. Il en résulte des conditions d'humidité élevée qui favorisent les troubles physiologiques de la laitue, en particulier la nécrose des extrémités extérieures des jeunes feuilles, connue sous le nom de "brûlure de la pointe". Cette thèse a observé les effets de différents traitements d'éclairage et de vitesses d'air sur la laitue à différents niveaux d'humidité comme moyen d'atténuer les effets délétères de l'humidité élevée. La laitue romaine (*Lactuca sativa*, cv. Teton) a été cultivée en hydroponie dans trois chambres de croissance sous trois humidités relatives (HR) constantes: 70%, 80% et 90%. Dans chaque chambre, trois traitements d'éclairage ont été testés: lumière blanche (WL), lumière blanche complémentée par du rouge lointain (WFR), et ambre large complémenté par du rouge lointain (WA). Chaque plateau de croissance comprenait d'un système de mouvement d'air construit à l'aide d'un tube de polyéthylène attaché à un ventilateur de conduit pour fournir deux zones de vitesse d'air: basse ($0,5 \text{ m s}^{-1}$) et haute ($1,0 \text{ m s}^{-1}$). Dans le système de mouvement d'air, la vitesse moyenne de l'air ressentie par les plantes variait de $0,23$ à $3,0 \text{ m s}^{-1}$. La réduction de la brûlure de la pointe a été très efficace, même lorsque l'humidité relative était à 90 %, car le mouvement de l'air a permis à la transpiration de rester très fonctionnelle tout en prévenant les troubles physiologiques liés à la diminution de la réponse stomatique dans des conditions prolongées d'humidité élevée. Les résultats indiquent que l'ajout de rouge lointain aux longueurs d'onde de l'ambre peut orienter le développement des plantes vers l'expansion des feuilles, en particulier dans des conditions d'humidité élevée. Le rouge lointain a probablement amélioré l'assimilation sous un faible rapport Rouge:Bleue (0,15), avec une amélioration de la conductance stomatique et une assimilation nette de CO_2 comparable à celle du WA et du WL, soutenant les recherches antérieures sur les effets promoteurs de la lumière bleue sur la régulation stomatique ainsi que les interactions synergiques potentielles avec le rouge lointain. Les résultats de cette étude indiquent que les systèmes de mouvement d'air

fournissant des vitesses d'air de 0,5-1,0 m s⁻¹ au niveau des plateaux sont efficaces pour éliminer les brûlures de la laitue à une humidité relative allant jusqu'à 90 % sans déclencher de réactions de stress supplémentaires. Les avantages du mouvement de l'air offrent également aux producteurs la possibilité de modifier les spectres d'éclairage afin d'optimiser davantage la croissance des plantes sans se préoccuper des effets négatifs d'une humidité élevée.

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Contribution of Authors

In adherence with the McGill University guidelines for a manuscript-based thesis, this section outlines the contributions made by the candidate and co-authors in completing this work. Jordan Wong is the principal author of all the chapters in this thesis, supervised by Dr. Mark Lefsrud from the Department of Bioresource Engineering, McGill University, Sainte-Anne-de-Bellevue, Québec, Canada.

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List of Abbreviations and Acronyms

Analysis of Variance – ANOVA

Blue Light – B

Chlorophyll – Chl

Controlled Environment Agriculture – CEA

Carbon Dioxide – CO₂

Daily Light Integral – DLI

Electroconductivity – EC

Far-red Light – FR

Green Light – G

Horizontal Air System – HAS

Light Compensation Point – LCP

Light Saturation Point – LSP

parts per million – ppm

Photosynthetic Photon Flux Density – PPFD

Potential of Hydrogen – pH

Red Light – R

Relative Humidity – RH

Specific Leaf Area – SLA

Ultraviolet Light – UV

Vapor Pressure Deficit – VPD

Vertical Air System – VAS

Chapter 1: Thesis Motivation

This chapter presents the background and rationale for the development of the research project. Research objectives for this study are stated.

1.1 Thesis Motivation

Lettuce (*Lactuca sativa*) is a leafy green vegetable widely consumed for its crisp texture, mild flavor, and nutritional value (Das & Bhattacharjee, 2020). The crop contains a significant amount of edible biomass as well as several essential minerals, vitamins, and fiber, contributing to its high demand among consumers (Weber, 2016). In recent years, lettuce has become a popular crop for large-scale commercial production due to its relatively short growth cycle, ease of cultivation, and high yield per plant (Das & Bhattacharjee, 2020). The crop is well suited to hydroponic cultivation, making it especially suitable for controlled environment facilities like greenhouses that frequently utilize this growing system. From 2018 to 2022, lettuce production in Canadian greenhouses increased by 278.1% from 7 million to 19.6 million kg (Statistics Canada, 2025). To obtain high quality produce intended for sale, it is important for growers to optimize the growing parameters of lettuce within their farms. One such parameter that will be explored in this study is humidity.

Humidity is an important environmental factor that influences the gas exchange processes of plants, notably transpiration (Spomer & Tibbitts, 1997). A common measure of humidity is relative humidity (RH), which measures air moisture relative to its moisture-holding capacity at a specific temperature (Grange & Hand, 1987). Higher RH levels (~85% RH) are conducive to greater crop yields by means of reducing stomatal resistance, allowing for corresponding increases in gas exchange processes such as photosynthesis and transpiration (Tibbitts & Bottenberg, 1976). This increase directly assists in facilitating calcium transport within the xylem via transpirational mass flow, preventing or mitigating 'tipburn,' exhibited by brown and wilted outer leaves; this disorder is particularly evident when RH is too low and plants undergo moisture stress, leading to reduced rates of transpiration that negatively impact

yield parameters such as fresh mass (Barta & Tibbitts, 2000; Farhangi et al., 2023; Grange & Hand, 1987).

Despite the benefits of higher RH levels, too much moisture in the environment can detrimentally affect plant growth (Spomer & Tibbitts, 1997). While RH levels of 80% upwards lead to greater rates of stomatal opening, this increases plant vulnerability to microbial infections, such as *Bremia lactuatae*, a common downy mildew pathogen (Su et al., 2004). While it has additionally been demonstrated that plant growth rates increase with increasing RH, the effect of increased transpiration rates may not be sufficient to overcome the difficulty in transporting calcium within the xylem due to the rapid growth rate of biomass, leading to tipburn. It may be that excess humidity can begin to slow transpiration instead of encouraging it, leading to more severe cases of tipburn (Frantz et al., 2004). Consequently, it is vital to maintain humidity within an ideal range to promote rapid plant growth without compromising on plant quality.

Controlling RH is not an easy task in a CEA environment, particularly in enclosed spaces such as small-scale plant factories or shipping container farms. By design, these facilities cultivate lettuce at very high densities within relatively small growing areas. While this is beneficial from both an economic and efficiency perspective, humidity control in these environments may prove difficult as lettuce is a short cycle plant that rapidly increases in biomass under electrical lighting conditions. The holistic effect of all the lettuce plants transpiring together can elevate RH to very high levels, over 90 % RH, potentially overwhelming HVAC systems that are not properly sized for cooling and dehumidification (Graamans et al., 2017). Crop quality is reduced as a result, indicating a need to explore alternative approaches to maintain high yield and quality of lettuce during periods of heightened RH.

Two common techniques utilized in CEA include blowing air at different velocities directly on the center of the plant to promote transpiration, while another more passive technique is to modify different aspects of the lighting spectrum used to grow the plants, such as spectral distribution or lighting intensity (Ahmed et al., 2020b). These methods have varying degrees of effectiveness depending on how they are implemented in the CEA facility. Therefore, more research is required to evaluate their effectiveness on lettuce growth, specifically in high

RH environments without suffering the detrimental effects of high RH nor compromising on factors that influence yield, food quality, or safety for indoor-grown crops.

1.2 Research Objective

Lettuce cultivated in high humidity environments may suffer from reduced transpiration rates, increased susceptibility to pathogenic infections, and tipburn, among other concerns. This study aims to determine how the effects of different lighting and air velocity treatments combined with elevated humidity levels influence the growth and development of lettuce cultivated in a controlled environment.

Chapter 2: Literature Review

2.1 Controlled Environment Agriculture (CEA) Overview

The population of the world is projected to peak at just over 10 billion individuals by the end of the century, according to median estimates by the United Nations (DESA, 2024). As such, the rate of food production will need to increase dramatically to adequately meet the nutritional needs of the burgeoning population. Production can be increased by expanding existing farmland into new regions or attempting to intensify production in existing farmland, but these options are not sustainable solutions in the long term. Figure 2.1 shows the increase in crop water consumption from 2000–2050 if agricultural intensification of existing farmlands continue (Pfister et al., 2011). Unsustainable farming practices can lead to heavy burdens on local water supplies, and intensification of agriculture will not be sufficient to meet the nutritional needs of the world population (Pfister et al., 2011).

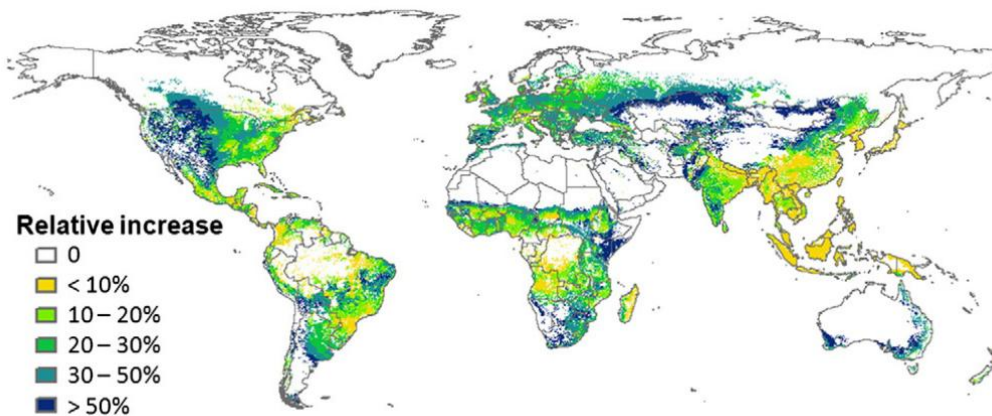


Figure 2.1 – Relative increase of crop water consumption in 2050 compared to 2000 for agricultural intensification (Pfister et al., 2011)

Agricultural expansion will become less viable as potential farmland will instead be dedicated to other services to support population growth. The loss of arable land is tied to the increase in urban areas as more people move from rural areas into the cities; approximately 50% of the world population in 2020 live within urban areas, and this is predicted to increase to 70% by 2050 (Hossain et al., 2020; Trewavas, 2001). The allotment of land towards urban areas

instead of farmland is especially troublesome in developing countries, where food production needs to be the most secure. Regions in India, Asia, and sub-Saharan Africa are predicted to be the hotspots of population growth in the coming decades, yet it is estimated that from 2000 to 2030, these countries will lose 30-60 MHa of farmland due to urbanization pressures (Döös, 2002). Developed countries are not immune either, as the United States is projected to lose 7.3 MHa by 2040 to urbanization centered in southern regions like California and Texas that are essential for food production (Xie et al., 2023). This decrease in available farmland is exacerbated by degradation of soil due to unsustainable practices, leading to 25% of soil used for agricultural and livestock production activities being rendered unusable as cropland due to a lack of essential nutrients (Hossain et al., 2020). The increasing severity of climate change will create more unpredictable weather patterns that will harshly impact the productivity and availability of farmland that has historically relied on stable climates (Graham, 2024). To meet the needs of the rising population, current agricultural practices need to shift towards a more sustainable approach that can provide greater yields to meet the nutritional demands of a rising population, be more land and resource efficient, and be less susceptible to the effects of climate change.

2.1.2 Controlled Environment Agriculture (CEA)

Controlled environment agriculture (CEA) offers a solution to food shortages by cultivating crops indoors, allowing for food to be grown year-round while remaining relatively insulated from external environmental conditions. Types of CEA facilities range from greenhouses, shipping container farms, and even massive multi-level plant factories, all of which can attain higher crop output per unit of land (Gómez et al., 2019). CEA facilities have the unique advantage of being able to expand crop cultivation upward instead of horizontally, a stark contrast to traditional agricultural techniques that are limited by land expansion. This upward growing strategy is coupled with the use of unique growing systems and techniques that allow growers to boost both the yield and quality of their crops while conserving resources. For instance, many vertical farms employ a growing system known as hydroponics, a soilless method of plant cultivation where a nutrient solution is used in combination with a growing

medium such as rock wool or perlite to hold the roots in place. Hydroponic systems can be precisely controlled to dictate the nutrients supplied to plants, allowing growers to fully quantify the nutritional needs of their crops to obtain the highest yields (Son et al., 2020). In addition to growing systems, CEA facilities maintain a high degree of control over environmental conditions in the facility, such as temperature, humidity, and lighting. Control of these factors allow these facilities to continue producing food even when outside conditions are not suitable for plant growth, allowing for consistent products year-round (Benke & Tomkins, 2017). All these factors highlight that vertical farms can produce much more biomass per square meter than traditional agricultural techniques. Gargaro et al. (2023) determined the yield of CEA lettuce to be double that of field grown lettuce (3.68 kg m^{-2} vs 1.87 kg m^{-2}).

Within the CEA industry, some of the most popular crops grown include leafy greens such as lettuce, tomatoes, herbs, microgreens, and strawberries (Benke & Tomkins, 2017). This study is primarily focused on optimization parameters for lettuce cultivation in controlled environments, acknowledging that many current solutions and techniques in CEA are applicable to a wide variety of crops.

2.1.3 Optimization for Lettuce Growth

CEA facilities go to great lengths to optimize lettuce growing conditions. Notable environmental parameters include humidity, lighting, air velocity, temperature, and carbon dioxide, with each parameter having varying impacts on the growth parameters of the final product (Ahmed et al., 2020b). It is imperative for growers to account for every condition and their potential interactions to protect their plants against diseases and disorders, while at the same time not neglecting one parameter over the optimization of another. An improperly balanced environment can stress plants and increase their susceptibility to different disorders, chief of which is tipburn for leafy greens like lettuce (Cox et al., 2015).

2.1.3.1 Tipburn

Tipburn is a physiological disorder prominent in lettuce cultivated in controlled environments whereby outer leaves brown and wilt (Figure 2) (Cresswell, 1991). While there are multiple factors that can potentially cause tipburn, the general consensus is that nutrient transport under rapidly growing lettuce is less efficient, leading to nutrient deficiencies at the outer edges of the plant (Barta & Tibbitts, 2000). Optimization of one of multiple environmental parameters for lettuce growth, such as lighting or temperature, can therefore increase risk of tipburn (Sago, 2016). As lettuce plants grow larger, the head develops and becomes more enclosed, potentially resulting in reduced transpiration rates due to increases in the leaf boundary layer resistance (Frantz et al., 2004).

Calcium is a vital nutrient that is primarily transported through the xylem via mass flow processes such as transpiration (Caplan, 2018). It is vital for cell wall strength and membrane integrity, and low concentrations of this nutrient can result in weaker cell structure (Caplan, 2018). Developing leaves undergoing high rates of expansion have high calcium demand, yet transport of nutrients to these outermost edges can be less efficient due to insufficient transpiration rates (Sago, 2016). Consequently, laticifer cells in these regions are weakened to the point of rupture and/or collapse, causing necrosis of plant tissue and therefore tipburn (Sago, 2016). While calcium may not be the primary nutrient affecting this change, several studies have demonstrated a correlation between reduced calcium concentrations in leaves and greater incidents of tipburn (Barta & Tibbitts, 2000). Mechanically reducing transpiration rates of plants via plastic sheaths can reduce calcium concentrations in young leaves and cause greater instances of tipburn, indicating that the disorder is at least heavily influenced by local calcium concentrations (Barta & Tibbitts, 1986).



Figure 2.2. Tipburned lettuce

Control of tipburn is difficult due to the many reasons that might be underlying the disorder (Frantz et al., 2004). Several studies propose the introduction of additional calcium into the growth cycle to overcome poor transpiration rates that reduce nutrient delivery to the outermost edges, as it has been demonstrated previously that tipburned plants have lower calcium concentration (Barta & Tibbitts, 1986). A study by Cresswell (1991) proposed circumventing transpiration as the main driver of nutrient uptake through root pressure, which normally develops at night as roots absorb more water than it is lost from the top, creating a gradient that is maintained by increased nutrient uptake. While tipburn was successfully reduced, it was not determined if this was due to an increase of average calcium concentration or improvement of calcium distribution in the leaf (Cresswell, 1991). Another study looked at foliar spraying of calcium solution throughout the growth cycle, which reduced tipburn incidence and increased calcium concentration in young leaves (Corriveau et al., 2012). Development of tipburn-resistant cultivars are being explored, although this would only be a first line of defense that would have to consider the rapid growth rates that lettuce cultivated in CEA can achieve (Koyama et al., 2011). In most CEA applications, the most common form of tipburn control is increasing transpiration rates of plants via elevated air movement over the plants. The subsequent effect is reduction of the boundary layer resistance to gas exchange

processes like transpiration, allowing growers to promote transpiration throughout their facility (Kitaya et al., 2004).

For leafy green production in CEA, controlling tipburn has two notable benefits: the first is that less tipburned plants simply mean a greater quantity of marketable crop. The second is that control of tipburn allows growers to increase the growth rate of plants through higher levels of light and CO₂ enrichment. These improvements would increase the yield of plant quality without suffering the deleterious effects of tipburn that would render them otherwise unmarketable. The following sections discuss the importance of each environmental parameter that needs to be controlled in CEA, describing effects on the physiological processes and morphological development of lettuce.

2.2 Humidity

Humidity describes the amount of water vapor in the air, generally defined as RH (Spomer & Tibbitts, 1997). RH describes the content of air moisture relative to its moisture-holding capacity, i.e. saturation, at a given temperature (Spomer & Tibbitts, 1997). This ratio can be derived from several measures of moisture content in the air as shown in Table 2.1 (Grange & Hand, 1987).

Table 2.1. Common measures of moisture quantity

Definition	Quantity	Units
Absolute Humidity	Mass of moisture per volume of air	kg m ⁻³
Specific Humidity	Mass of moisture per mass of air	kg kg ⁻¹
Vapor Pressure	Partial pressure of water vapor	kPa
Dew Point Temperature	Temperature where vapor pressure of air is equal to saturation vapor pressure	°C

While RH is a helpful measure of gauging humidity for growers, it does not accurately capture the moisture-holding capacity of air at different temperatures since saturation levels vary with temperature (Spomer & Tibbitts, 1997). RH does not accurately characterize the flow of moisture in a system in processes such as transpiration and evaporation, in which vapor

pressure deficit (VPD) is a much more suitable measure (Grange & Hand, 1987). VPD describes the difference between the saturation vapor pressure and the actual vapor pressure and is independent from temperature (Wollaeger & Runkle, 2016). A high VPD (>1.0 kPa) indicates that the air is still far from saturation, while a low VPD means air is close to saturation (Wollaeger & Runkle, 2016).

Transpiration is the movement of water and nutrients from the leaf to the surrounding environment due to the vapor pressure difference (Gómez et al., 2019). Originally described by Penman and further expanded upon by Monteith to include specific plant characteristics, the energy balance of a transpiring leaf can be modeled using Equation 2.1, including net radiation (R_{net}), sensible heat exchange (H) and latent heat exchange (λE) (Monteith, 1965).

$$R_{net} - H - \lambda E = 0 \quad (2.1)$$

Of particular interest is the latent heat exchange term, or rather transpiration itself, which is directly modulated by the VPD between the plant and the air as shown in Equation 2.2. LAI is leaf area index, r_s is stomatal resistance, r_a is aerodynamic resistance (or leaf boundary layer resistance), χ_s is vapor pressure at the transpiring surface and χ_a is the vapor pressure of the surrounding air (Graamans et al., 2017).

$$\lambda E = LAI \cdot \lambda \cdot \frac{(\chi_s - \chi_a)}{r_s + r_a} \quad (2.2)$$

Based on Equation 2, a larger VPD between the leaf and air will lead to greater increases in transpiration, while a smaller VPD will decrease transpiration. Therefore, an optimized VPD can induce greater rates of transpiration to allow for better nutrient transport and water flow throughout the plant (Spomer & Tibbitts, 1997). This is especially pertinent for calcium, as this fortifying nutrient is absorbed at the roots and transported through the xylem via transpiration-induced water flow (White & Broadley, 2003).

It is expected based on Equation 2.2 that a larger VPD would lead to corresponding increases in transpiration and therefore plant performance. However, studies tend to report

mixed results on the ideal VPD for plant growth depending on the plant. A review by Hoffman for 26 crops (ranging from fruits, leafy greens, legumes, cereals, etc.) reported the greatest improvements in plant growth from VPDs of 1.8 to 1.0 kPa with no additional benefits when pushed to 0.3 kPa (Hoffman, 1979). Tibbitts reported improved lettuce growth rate at 85% RH (0.35 kPa) compared to 50% RH (1.17 kPa), with reduced stomatal resistance at 85% RH (Tibbitts & Bottenberg, 1976). Shamshiri et al. (2018) similarly reported good growth for VPD ranging from 0.3-1 kPa for tomato cultivation. van de Sanden (1985) observed no significant differences in both growth parameters and gas exchange between cucumbers grown at 90% (0.32 kPa) or 60% RH (1.27 kPa). Therefore, although difficult to precisely pinpoint an optimum VPD for plant growth, 0.3-1.0 kPa seems to be the range to prevent overt reduction in plant function (Hoffman, 1979). Increasing or decreasing humidity within this range has relatively minor effects on plant growth, and this holds true for across a wide range of commonly grown agricultural crops (Grange & Hand, 1987; Hoffman, 1979).

Elevated humidity (>85% RH) or lower VPD (<0.1–0.2 kPa) reduces performance of physiological processes (Spomer & Tibbitts, 1997). Lowered transpiration rates are the primary effect, potentially causing tipburn as the outermost edges of leaves furthest away from the roots may not receive the necessary calcium to strengthen cell walls (Clarkson, 1996). Leaves within the meristem suffer similarly, as the relatively enclosed environment is non-conductive to transpiration due to clustering of leaves that promote stagnation of the leaf boundary layer (Frantz et al., 2004). This effect can be seen in Equation 2.2, where an increase in the air resistance (r_a) will directly reduce transpiration rate. A study by Barta and Tibbitts (1986) supports this observation, where it was reported enclosed developing leaves are highly likely to have reduced transpiration rates and therefore be tipburned. Deleterious effects of leaf enclosures extend to dense, tightly packed canopies in a highly humid environment that are common in CEA facilities (Barta & Tibbitts, 1986). Notably, RH greater than 90% can drastically promote the growth of fungal infections and *Bremia lactucae*, where open stomata provide an entry point for infections to enter and propagate (Su et al., 2004).

While optimal humidity is desirable for plant growth, it is both difficult and expensive to control for this parameter in CEA facilities, due to the balance of dehumidification and

temperature control being an energy-intensive endeavor that can be complicated by the unique climate that the facility is located in (Kozai & Niu, 2016). The exchange of indoor air with outdoor air using vents and fans is a common method for humidity control and usually works best with cool, dry air climates. Hot and humid climates will struggle to maintain both humidity and temperature, and internal humidity can hover upwards of 80–85% (Gómez et al., 2019). The energy input into the system from growing crops in a dense food production environments like greenhouses and plant factories can overwhelm HVAC systems, making dehumidification all the more difficult if ventilation systems are not designed or implemented properly (Graamans et al., 2017). Growers often face elevated humidity levels, which can lead to reductions in plant physiological function and increased susceptibility to disorders or diseases (Kozai & Niu, 2016). Low humidity levels are also possible and are associated with reduced plant growth parameters, as plants under water stress close their stomata to prevent further moisture loss (Ahmed et al., 2020b; Chia & Lim, 2022).

To mitigate the damaging effects of heightened humidity on plant growth and physiological processes, other environmental factors common in CEA such as air velocity and lighting can be optimized to compensate or mitigate reduced performance in physiological processes.

2.3 Air Velocity

Air velocity is a primary method used to control tipburn in CEA facilities. Blowing air on growing plants most directly impacts the leaf boundary layer (r_a), a thin layer of air that surround each leaf of a plant which acts as a resistance to gas exchange processes such as transpiration and photosynthesis (Ahmed et al., 2020b). The thickness of this layer can vary based on microclimate conditions in the canopy, but heightened humidity levels or temperature can potentially thicken this layer (Barta & Tibbitts, 1986). A thicker boundary layer provides greater resistance to gas exchange, as well as trapping heat and moisture in the microclimate surrounding the plant (Monteith, 1965). In response, a greater proportion of stomata close to reduce moisture loss, leading to a reduction of overall stomatal conductance and gas exchange (Fanourakis et al., 2020). This effect is particularly concerning as lettuce is a hypostomatous

plant, meaning that decreases in stomatal conductance have a much greater negative impact on gas exchange as opposed to amphistomatous plants (Drake et al., 2019). Therefore, blowing air on leaves is essential to ensure proper gas exchange can occur by acting as a microclimate regulator, preventing elevated temperatures and humidity in the canopy.

While minimizing the impact of the leaf boundary layer is ideal for encouragement of gas exchange, it is imperative to consider both the speed and distribution of air velocity (Zhang et al., 2016). Both excessively low and high air velocities can negatively impact photosynthesis rates (Ahmed et al., 2022). Very low air velocities can increase the boundary layer thickness, while high air velocities can result in stomatal closure to limit water loss due to the wind-encouraged evaporation (Ahmed et al., 2020b).

High air velocities can induce defense responses in plants that lead to reallocation of resources away from biomass accumulation (Chehab et al., 2008). Studies have reported the ideal velocity for maximum gains in gas exchange for lettuce lies somewhere between 0.3-1 m s⁻¹ throughout the growth cycle (Ahmed et al., 2020a; Kaufmann, 2023; Kitaya et al., 2004; Lee et al., 2013; Sohn et al., 2023). These values are normally reported as an instantaneous moment in time or as a constant state value, with changes in air velocity at the plant level normally unreported. These changes may be significant for plant growth and physiological processes, as a given plant can experience varying velocities from the same air movement system throughout its life cycle until maturity (Zhang et al., 2016).

Directionality and uniformity of air movement is vital (Kaufmann, 2023). Microclimate conditions in just one tray can vary drastically depending on the air movement system that is utilized, therefore it is desirable to create a homogenous system that is able to deliver similar air velocities to all plants in a tray to have consistent plant quality (Zhang et al., 2016). Studies have reported on different air movement systems using computational fluid dynamics (CFD), a method of simulating complex physical phenomena in a CEA facility by allowing users to simulate all environmental parameters that can affect air flow, such as lighting, temperature, pressure, etc. with reasonable accuracy (Fang et al., 2020; Peiro et al., 2020; Sohn et al., 2023; Zhang et al., 2016). Models are designed within the CFD software using the expected operating conditions in real life, then tested in existing CEA setups to determine the actual effectiveness of

the air movement system on plant growth parameters. Zhang et al. (2016) reported a design for an air movement system using a perforated air tube as shown in Figure 2.3, with the greatest error between the measured air velocity of the simulated and the CEA system being 12.2%. A study by Kaufmann (2023) implemented Zhang's system for lettuce growth, observing reductions in both instances and severity of tipburn.

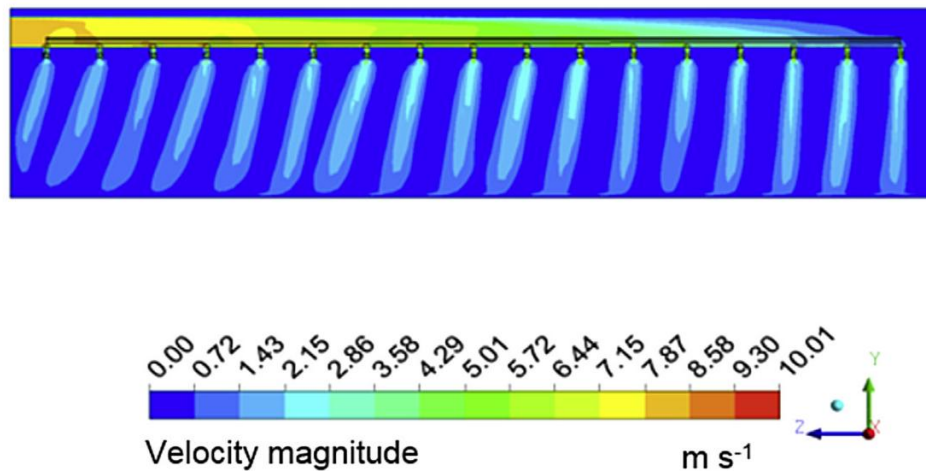
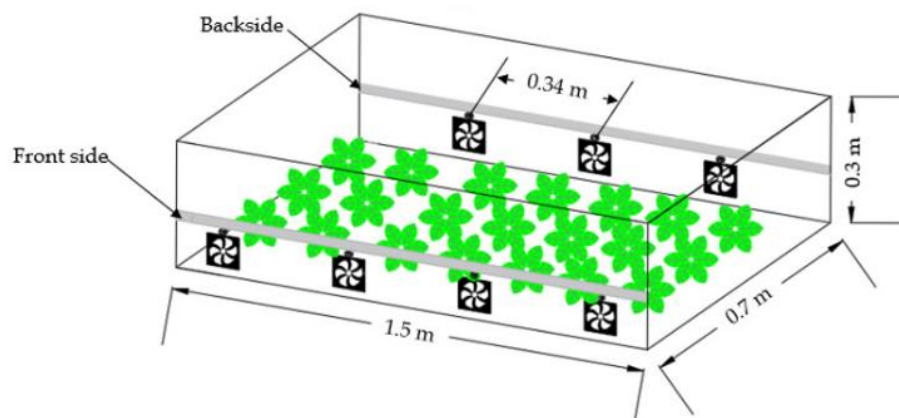


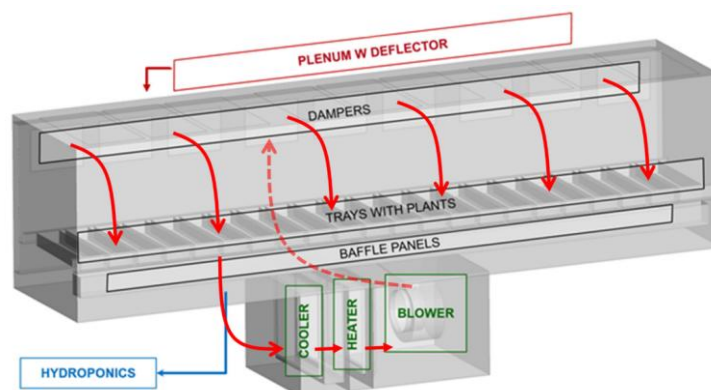
Figure 2.3. CFD Model of a perforated tube air movement system (Zhang et al., 2016)

Several different types of air movement systems exist in CEA facilities and have been reported in literature, but the most commonly used are vertical or horizontal systems, where the air source is placed either above the trays or at the ends or sides of the tray, respectively (Fang et al., 2020). Vertical air systems (VAS) utilize either ventilation ducting or deflectors to deliver air to plants, while horizontal air systems (HAS) place fans either along the growth trays or at the two opposite ends (Figure 4). Both are useful systems to reduce tipburn and promote gas exchange, and several studies have reported positive results from the inclusion of either VAS or HAS in their growth systems for leafy greens (Ahmed et al., 2020a; Kaufmann, 2023; Lee et al., 2013; Peiro et al., 2020). In a direct comparison between the two, VAS appear more effective as it allows for more uniformity of airflow throughout the tray. VAS better target meristems that are more vulnerable to tipburn and boundary layer stagnation (Kaufmann, 2023). Vertical ducts can be positioned next to or between lighting fixtures, allowing simultaneous temperature control of the plant canopy microclimate against the heating effects of the lighting treatment.

HAS requires airflow from the fan to travel through the edge plants into the interior row plants, meaning air flow distribution is disproportionately skewed higher towards the edge plants (Sohn et al., 2023). This effect is further exacerbated the thicker and denser the plant canopy is, meaning a greater crop resistance to air flow that cannot easily be controlled or modeled (Sohn et al., 2023). Center plants are in more danger of tipburn later in the growth cycle when they are more vulnerable due to rapid leaf expansion from surrounding plants (Frantz et al., 2004). Edge plants may be negatively affected as well due to high velocity levels at the edge, further reducing the overall performance of the growth tray (Zhang et al., 2016). For this study, a VAS was employed, following methodology described previously by Kaufmann (2023).



(a)



(b)

Figure 2.4. Examples of two airflow systems for plant growth: (a) horizontal air system (Ahmed et al., 2020a) and (b) vertical air system (Peiro et al., 2020)

A knowledge gap in CEA literature is plant morphogenic responses to mechanical stimulation over time, referred to as thigmomorphogenesis (Graham & Wheeler, 2017). Plants have evolved mechanisms to respond to mechanical stimulation or perturbations in their natural environment, such as from winds or storms (Chehab et al., 2008). Thigmomorphogenic responses normally effect morphology of the plant over time, such as decreases in shoot elongation coupled to an increase in radial expansion, strengthening of existing structures, and even shorter and more compact plants (Chehab et al., 2008). Responses can additionally fall under cellular signaling pathways, leading to adjustments in chlorophyll and hormone levels (Chehab et al., 2008). Studies on thigmomorphogenesis have been conducted in controlled environments on plants like tomatoes and bell peppers with varying results. Mitchell (2012) observed that shaking tomatoes 30 seconds once or twice daily led to shorter internodes and leaves, lower biomass, and stronger stems. Similarly, Graham and Wheeler (2017) mechanically stimulated bell pepper plants during vegetative growth, reducing all growth parameters but increasing chlorophyll (SPAD) content. The same study also observed that shorter plants generally had better volume use efficiency (Graham & Wheeler, 2017). Overall, thigmomorphogenesis is a complex mechanism controlled by a variety of different genes and signaling pathways, with limited research reported for lettuce response, let alone in CEA (Graham, 2024). While it is almost certain that any air movement system utilized will impact the morphology of the plant via thigmomorphogenic mechanisms, it is difficult to isolate its effects to the complex interaction of other environmental variables like air boundary layer, lighting, humidity, etc. in CEA (Chehab et al., 2008).

2.4 Lighting

Lighting is a critical driver of plant development, playing a key role in gas exchange processes such as photosynthesis and transpiration (Sager & McFarlane, 1997). Significant research has been performed to evaluate the effects of different aspects of lighting, most notably spectral composition, intensity, and duration (Gómez et al., 2019; Jeong et al., 2025; Sager & McFarlane, 1997; Sago, 2016). Changing any of these parameters, such as modifying the

wavelength ratio or increasing the intensity, can have a major impact on the final quality and yield of the crop (Boucher et al., 2023; Hogewoning et al., 2012). As such, it is imperative that each of these parameters be optimized and balanced with each other to achieve high plant yield and quality.

2.4.1 Spectral Composition

Spectral composition is the distribution of light energy across the range of wavelengths known as photosynthetically active radiation (PAR) (Wu et al., 2025). This is the region of the visible spectrum between 400 and 700 nm that plants respond to and are able to utilize to drive photosynthesis (McCree, 1971). While formerly thought to strictly cut off before 400 nm and after 700 nm, recent research has demonstrated that plants make use of both ultraviolet (280–400 nm) and far red (700–800 nm) wavelengths, expanding the definition of PAR to 280–800 nm (Zhen & Bugbee, 2020a, 2020b). Figure 2.5 shows the original range of PAR as determined by McCree (Liu & van Iersel, 2021; McCree, 1971). This section will discuss each wavelength range and recent research that has been conducted on its effects on different plant processes. Spectral effects may differ from crop to crop, but their influence on crop morphology and physiology is generally applicable across different crops.

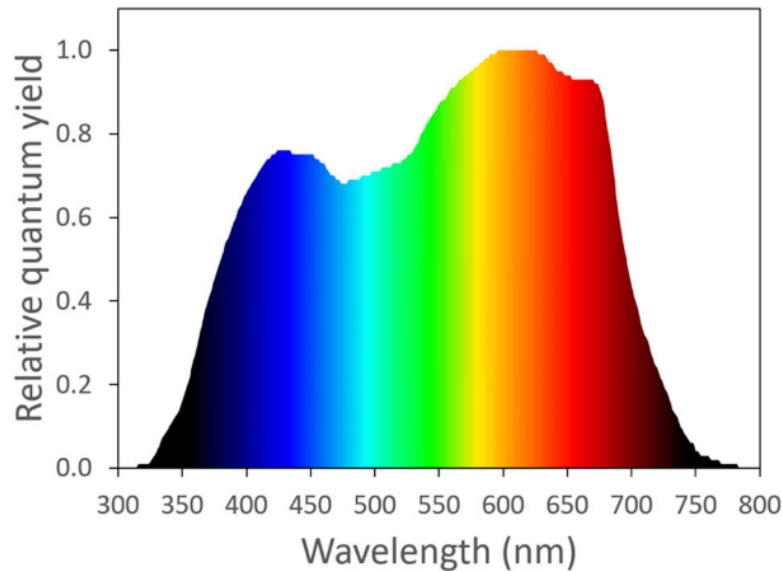


Figure 2.5. Normalized action spectra of max quantum yield of CO assimilation (Liu & van Iersel, 2021; McCree, 1971)

While monochromatic wavelengths tend to have relatively defined effects on different plant physiological parameters, it has been well documented in literature that the interaction of different wavelengths has an additional effect on plant development. This is especially true in regard to photosynthesis, as balancing excitation of the constituent parts of the photosynthetic apparatus (photosystem I (PSI) and photosystem II (PSII)) influences photosynthetic efficiency (Emerson et al., 1957; Hogewoning et al., 2012; Wu et al., 2025). Characteristics of monochromatic light are important to consider, but interaction effects between different lighting spectrums are just as essential (Demotes-Mainard et al., 2016; Emerson et al., 1957; Heraut-Bron et al., 1999). Discussion of each spectrum will include effects of both monochromatic and interaction effects with different spectra.

2.4.1.1 Red light

Red light (R) ranges from 600–700 nm and is a major effector on plant growth and development (Izzo et al., 2020). It contains the most efficient wavelengths for driving photosynthesis. In monochromatic studies, both McCree (1971) and Inada (1976) observed the

highest quantum yields at roughly 620–660 nm, and this has been well correlated with additional spectral analyses by Evans (1987) and Hogewoning et al. (2012). R promotes stem elongation, leaf expansion, biomass accumulation, and other light-mediated responses of plants (Sager & McFarlane, 1997). Monochromatic R is an ineffective driver of plant growth (Wang et al., 2016). Plants grown under monochromatic R suffer from an under stimulated PSII and therefore lower net assimilation, with plants becoming more susceptible to osmotic stress with reduced stomatal conductance (Savvides et al., 2012; Wang et al., 2016). Early exposure to monochromatic R in developing leaves can permanently damage parts of the photosynthetic system (Trouwborst et al., 2016). Plants then struggle to recover from this damage, reducing CO₂ assimilation and the performance of other associated leaf factors like stomatal conductance (Trouwborst et al., 2016). Despite high instantaneous performance as an individual wavelength in monochromatic studies, the long-term effects of monochromatic R on photosynthesis and plant growth may not be as beneficial (Trouwborst et al., 2016).

R is often combined with blue light (B) due to photons of both wavelengths being highly absorbed by photosynthetic pigments, based both on McCree's original study as well as later analyses of absorption spectra (Inada, 1976; McCree, 1971). While neither wavelength is effective when supplied monochromatically, together they synergistically combine to compensate for what the other lacks (Hogewoning et al., 2010; Wang et al., 2016). Substantial research has been performed on the effects of modifying the R:B ratio, with varying but overall positive effects on all plant growth parameters and processes (Wang et al., 2016). Several studies that tested the effects of monochromatic R found that the inclusion of B improved performance of lettuce physiological processes, with increases in photosynthetic capacity, rate, and stomatal conductance as the ratio of R:B decreased until 1, as shown in Figure 6 (Wang et al., 2016). Trouwborst et al. (2016) observed that the addition of B after initial exposure to monochromatic R for a ratio of 70% R to 30% B was able to reverse some of the damage done to photosynthetic capacity in cucumbers, indicating the necessity of both R and B wavelengths in combination. An additional review study by Bantis et al. (2018) supports the positive effect of B supplementation to R, noting improved plant biomass, plant height, stem diameter, and even increased protein and chlorophyll concentrations across commonly cultivated greenhouse

crops. Therefore, while the precise ratio of R: B will need to be dialed in by growers based on their specific needs, it has been well established that these two wavelengths in tandem can improve overall plant performance for all important parameters.

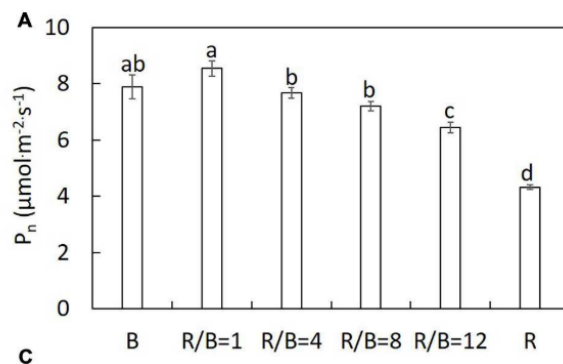


Figure 2.6. Effect of red to blue (R:B) ratio on net photosynthetic rate for lettuce, where increasing proportion of B lead to significant improvements in photosynthesis compared to monochromatic R (Wang et al., 2016)

2.4.1.2 Blue light

Blue light(B, 400–480 nm) is a major driver of different plant processes and growth parameters, affecting aspects from plant growth, leaf expansion, stomatal regulation, and photosynthesis among others (Hogewoning et al., 2010). While monochromatic B does not have the highest quantum yield when compared to red, it still ranks second in terms of induction of photosynthesis as originally demonstrated by McCree and later confirmation by Inada (Inada, 1976; McCree, 1971). B is often introduced into spectral distributions with R to compensate for the previously discussed deficiencies of monochromatic R. In studies where B is introduced in a lighting spectrum with R, regardless of the ratio between R:B, photosynthetic performance improved with increasing ratio of B up to 50%, with diminishing gains past this point (Ahmed et al., 2020b). This combination can lead to enhanced transpiration rates due to the positive effect of B on stomata, with a greater proportion of blue light increasing stomatal conductance in lettuce (Ahmed et al., 2020b).

Studies have observed that marginal addition of B is sufficient to significantly improve photosynthesis and growth parameters, with 7% blue enough to double photosynthetic capacity (A_{max}) and 5% improving flowering and fruit yield in peppers (Hogewoning et al., 2010; Naznin et al., 2019). Lettuce plants grown under monochromatic B perform poorly compared to more balanced lighting spectra, with reduced stomatal density, root dry mass, and leaf thickness (Izzo et al., 2021). When compared to other monochromatic spectra like R, chlorophyll abundance is still high, and this can be correlated with photosynthesis under B still being relatively functional despite reduced efficiency (Hogewoning et al., 2010; Izzo et al., 2021). To that end, if the ratio of R: B is properly determined, marked improvements in plant performance can be achieved, with the bulk of photosynthetic needs met by just these two wavelengths (Wang et al., 2016). Further optimization can be achieved through supplementation of additional wavelengths (Wu et al., 2024).

2.4.1.3 Ultraviolet

The ultraviolet (UV) spectrum lies outside the visible spectrum on the opposite end of far-red and includes wavelengths from 10 to 400 nm (Verdaguer et al., 2017). This range is further subdivided into, in order of increasing energy, UV-A (315–400 nm), UV-B (280–320 nm), and UV-C (< 280 nm), where UV-A and UV-B are the primary components of natural sunlight (Kozai et al., 2016). Photons at these shorter wavelengths are at a much higher energy level, where overexposure can lead to stress and production of reactive oxygen species (ROS) (Chen et al., 2019). ROS can act as oxidizers of lipids and proteins, causing severe damage to cell structure and function (Chen et al., 2019). UV-B induces formation of pyrimidine dimers in nucleic acids, leading to irregularities in transcription (Kimura et al., 2004). Furthermore, there is ample support that all parts of the UV spectrum can have detrimental effects on photosynthetic parameters. An overexposure of UV-A overexposure negatively impacts PSII through reduced maximal quantum efficiency that lead to corresponding decreases in the electron transfer rate and photosynthesis (Vass et al., 2002). This may be attributed to a reduction in Rubisco or overproduction of ROS that inhibit production of PSII proteins (Takahashi & Badger, 2011; Verdaguer et al., 2017). For UV-B, its mere presence is enough to act

as a general suppressor of photosynthesis and plant morphology, decreasing the concentration of photosynthetic pigments and reducing leaf area to create more compact plants, respectively (Chen et al., 2019; Kataria et al., 2013; Vass et al., 2002).

Despite the potential deleterious effects of UV, plant growth fixtures in the past years have started including UV-A in low doses in their spectra to induce production of secondary metabolites like flavonoids and anthocyanins that improve the nutritional profile of the plant (Chen et al., 2019). These compounds act as a form of disease control, inducing protective defense responses in plants as well as killing microorganisms in the growth environment (Vanhaelewyn et al., 2020). At low intensities, for lettuce and tomatoes, UV-A and UV-B wavelengths act as signals for production of secondary metabolites or phenolic compounds that absorb UV to mitigate accumulation of ROS and free radicals (Chen et al., 2019; Prado et al., 2012). This production is coupled with increases in leaf area or biomass as a response to perceived stress, similar to artificial induction of shade avoidance syndrome using FR wavelengths (Chen et al., 2019; Tezuka et al., 1993). UV-B at low doses reportedly prevents intumescence, a physiological disorder characterized by abnormal outgrowths of cells on leaf, petiole and stem surfaces (Kozai et al., 2016) .

Damage caused by UV photons can be mitigated, as the degree of damage varies based on the duration, intensity, and specific UV wavelengths that are used (Kataria et al., 2013). The shortest wavelengths at UV-C will burn plants after exposure for just a few seconds, while the highest wavelengths at UV-A can be tolerated for longer periods of time at low levels (Chen et al., 2019). Negative impacts on photosynthesis are less severe at low amounts of UV-A, or at the very least UVA only begins to manifest photoinhibition close to saturation levels (Chen et al., 2019). One study by Chen et al. (2019) observed no distinct reductions in photosynthetic rates at lower levels compared to control treatments. UV-A is normally analyzed by itself or in combination with UV-B, with the effects of interactions between them or on longer wavelengths on plant morphology and photosynthesis varied at best.

2.4.1.4 Green light

Green light (G) is defined by the wavelengths between 500–580 nm and has historically been considered less important for photosynthesis compared to R and B due to chlorophyll's weak absorbance of green photons (Mitchell & Stutte, 2017). This assumption is based off the individual response of a leaf to G, yet more recent literature has demonstrated that G has much better penetration into lower canopy levels despite its low absorbance at the top level (Brodersen & Vogelmann, 2010). Even in McCree's founding work, despite a dip relative to other wavelengths, the absorbance of G from 0.60–0.82 indicated that the majority of photons were being absorbed (McCree, 1971). One potential explanation is that green photons that are not immediately absorbed are repeatedly reflected from chloroplast to chloroplast, increasing the chances for it to be absorbed (Kim et al., 2004). This penetrating effect can create a more uniform light distribution throughout the plant, aiding in light exposure for leaves that would not normally receive the more photosynthetically active red or blue photons. This extra source of lighting can potentially improve photosynthetic parameters such as gas exchange rates through improvement of stomatal conductance, demonstrating that the supplementation of G to a B and R spectra can be greatly beneficial for plant growth (Liu et al., 2017).

The penetrating effect of green photons into lower canopy levels cause G to act as a low-light signal (Wang & Folta, 2013). A lighting environment enriched with G is likely to be enriched in foliage and therefore deficient in more photosynthetically active wavelengths that are unable to penetrate as deeply (Wang & Folta, 2013). Consequently, G can act as a signaling mechanism that induces shade avoidance responses, where plants adjust their physiological mechanisms in order to compensate for reduced lighting conditions (Dou & Niu, 2020). This can manifest through elongated petioles, leaf hyponasty, or reallocation of biomass towards leaf expansion and stem support away from roots (Wang & Folta, 2013). Some research has suggested that G alone does not induce shade response, or rather that the shade avoidance responses are much more pronounced when augmented with photons from the far-red (FR) spectrum (Wang et al., 2015). FR is found in high amounts in shaded environments due to the reduction of the R:FR ratio when compared to canopy level, and there is some support that FR and G can synergistically combine to induce shade avoidance responses despite differing mechanisms (Kelly & Runkle, 2024; Wang et al., 2015). Similar to FR, there is some evidence that increasing

the fraction of G greater than 50% of the spectrum can have detrimental effects on growth parameters like dry mass and leaf area (Kim et al., 2004). This is hardly surprising, as overexpression of shade-avoidance traits can inevitably be more harmful to the plant in the long-term (Casal, 2013).

From a human perspective, G is beneficial because it makes it easier to identify any diseases or unusual coloration in leaves (Kim et al., 2004). Plants readily absorb B and R at the top level, resulting in plants grown under these wavelengths appear purplish gray. G can further be used in postharvest storage of lettuce to increase nutritional quality, such as antioxidants and secondary metabolites (Salehinia et al., 2024).

2.4.1.5 Amber light

Amber light (or yellow light) is defined from approximately 570–620 nm and can exude numerous effects on plant physiological development. While McCree's original study pointed out that R at 625 nm had the highest quantum yields, its performance was not significantly different from amber light at 575 nm (McCree, 1971; Wu et al., 2023). Unfortunately, despite the latent potential of amber wavelengths, prior research has only lightly touched upon wavelengths in this specific region. More commonly, because conventional definitions of G tend to go up to 600 nm, prior studies on the effects of G include amber wavelengths as well, without naming it directly. For instance, a study by Terashima et al. (2009) studied the effects of increasing photosynthetic photon flux density (PPFD) on quantum yield for a monochromatic G spectrum with a peak at 550 nm, concluding that at high PPFD G was able to drive photosynthesis more effectively than R. Notably, the green spectrum had a half bandwidth of ± 30 nm, meaning that the spectrum did include some amber wavelengths (Terashima et al., 2009). While this observation isn't a mandate on the effects of amber, it demonstrates that prior research on wavelengths in this region tend to only consider peak effects within the green spectrum without considering the wavelengths included within the broader bandwidth. Later research that aimed to test amber wavelengths more directly continue this problematic trend due to lighting technology limitations that led to varying bandwidths being tested. For example, a 10-nm bandwidth from a low-pressure sodium lamp (Britz, 2008), a 25-nm bandwidth from

monochromator-filtered light (McCree, 1971), and a 70–80-nm bandwidth with full-spectrum LEDs (Wu et al., 2021) have been reported. This large variation in the amber light spectrum creates challenges when evaluating the impact of amber light on plant production, and it hasn't been until recently that more in-depth research has been conducted on the potential of amber light supplementation (Reddy et al., 2024; Wu et al., 2021).

A more recent study by Wu et al. (2025) isolated the photosynthetic parameters of amber wavelengths as part of a study on the full PAR spectrum. The full spectrum was tested in 1-nm increments with a half bandwidth of 10 nm for each wavelength tested. For both green and amber regions (575 nm and 595 nm respectively), quantum yield was found to be the higher compared to the rest of the PAR spectrum as shown in Figure 2.7, lending credence to the theory that amber light shares penetrative properties with G that allow it to induce greater rates of photosynthesis (Wu et al., 2025).

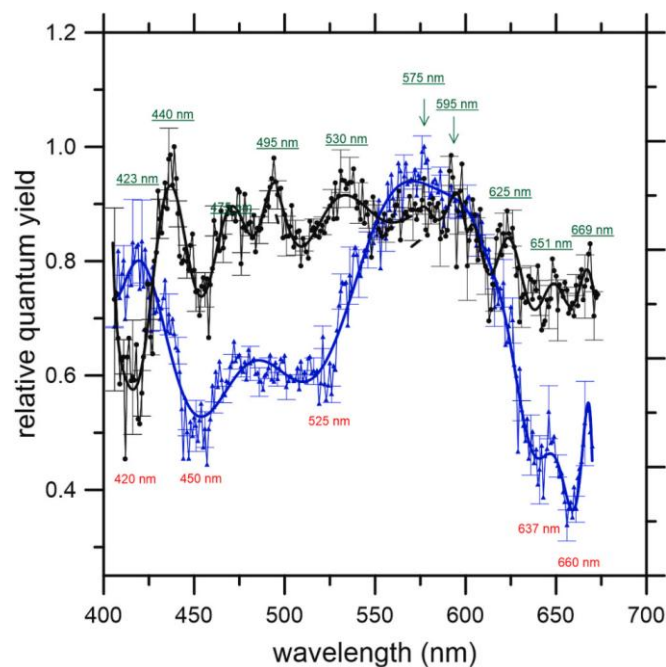


Figure 2.7. Quantum yield for photosynthetically active radiation (PAR) spectrum (Wu et al., 2025)

While the precise photosynthetic properties of amber remain yet to be fully explored, growth trials conducted under amber lighting have elucidated some of its effects on

physiological and morphological traits of crops throughout their growth cycle (Reddy et al., 2024; Trumpler et al., 2024). Current studies have explored the performance of monochromatic amber, supplementation of different wavelengths to amber wavelengths or vice versa, and wide amber (bandwidth ~ 80 nm) vs narrow amber (bandwidth ~ 20 nm). Reddy et al. (2024) compared the performance of monochromatic R, B, and wide amber (FWHM = 76.3 nm) at different intensities on lettuce, reporting the highest productivity in amber with 33% higher biomass accumulations than R up to $700 \mu\text{mol m}^{-2} \text{sec}^{-1}$. Above $800 \mu\text{mol m}^{-2} \text{sec}^{-1}$, growth rate was suppressed, and bleaching occurred only for amber above $1000 \mu\text{mol m}^{-2} \text{sec}^{-1}$ (Reddy et al., 2024). Another study with tomatoes reported similar findings, where wide amber light outperformed competing wavelengths in fresh mass and dry mass across different intensity levels (Wu et al., 2024). These findings indicate the potential of wide amber wavelengths in driving photosynthesis and plant growth.

Narrow amber drives high photosynthetic activity; however, additional wavelengths are necessary to maintain healthy plant growth (Trumpler et al., 2024). Monochromatic narrow amber alone lacks the wavelengths required to maintain morphologically healthy plants and performs worse in plant growth parameters compared to wide amber (Wu et al., 2023). When supplemented with B wavelengths, performance across the board improve dramatically, with positive results reported for tomatoes (Wu et al., 2023). Wide amber can receive similar benefits with blue supplementation, although the results can be crop dependent. Trumpler et al. (2024) noted that shifting the blue peak from 430 nm to 455 nm with broad amber lighting led to 40% less fresh mass for tomatoes, whereas only an approximate 5% reduction in fresh mass for lettuce.

Additional research is required to better characterize the full effects of amber wavelengths. Current literature draws similarities between amber and other wavelengths, with monochromatic amber (wide and narrow) performing better with supplementation of B just like R (Trumpler et al., 2024). Reduced performance at high intensities of amber indicates reduced photosynthetic efficiency, potentially due to an imbalance in the photosynthetic apparatus, although this remains to be explored in depth (Wu et al., 2024). Bandwidth seems to play an important role in the effect of amber light, although whether broad or narrow amber light has

better performance seems to depend on supplemental wavelengths (Wu et al., 2021). Despite conflicting data, the photosynthetic potential of amber light is well-established, with high potential for plant utilization if the lighting spectrum is optimized.

2.4.1.6 Far-red light

Far-red (FR) light, ranging beyond the visible spectrum from 700–780 nm, lies outside the range of what is considered traditionally considered PAR. Its importance has become increasingly evident in recent years as both an important signaling mechanism for plants as well as for enhancing photosynthetic response (Zhen & van Iersel, 2017). As a signaling mechanism, the amount of FR that a plant receives is a reliable indicator for when the plant is in shade (Kozai et al., 2016). In natural environments, plants perceive radiation from the sun using photoreceptors within photosynthetic pigments for specific wavelengths, including R, FR, B, and UV radiation (Park & Runkle, 2017).

Photosynthetic pigments will either reflect or absorb these wavelengths in different combinations, with R and B normally the most strongly absorbed wavelengths while FR is reflected or transmitted (Holmes & Smith, 1975). This leads to a normal daylight ratio between R and FR around 1.2 (Holmes & Smith, 1975). Spectral composition changes when the plant is under shaded conditions, and leaves deeper within the canopy will experience a greater proportion of FR as R is absorbed and FR is transmitted and reflected, reducing the R: FR ratio from 1.2 to 0.2 (Heraut-Bron et al., 1999).

A reduction in R: FR triggers a variety of shade avoidance responses that compensate for reduced lighting conditions (Casal, 2013). More resources can be allocated to strengthen stems as opposed to leaves, leading to stem elongation and thinner, yet wider leaves, with the goal of expanding potential leaf area of radiation capture as large as possible (Demotes-Mainard et al., 2016). Reduction in branching and lower allocation of biomass to roots follows this trend of resource reallocation towards expansion, and leaves can either preferentially expand towards areas with higher R:FR ratios away from neighbors, or exhibit hyponastic responses where the leaves grow up at a higher angle (Casal, 2013; Demotes-Mainard et al., 2016; Zhen & Bugbee, 2020b).

From a photosynthetic perspective, literature has well established that FR photons alone have poor photosynthetic activity, leading to the original misconception that FR photons have no impact on plant photosynthetic pathways (Inada, 1976; McCree, 1971; Wu et al., 2025). However, Emerson et al. (1957) demonstrated that FR synergistically interacts with shorter wavelengths in the visible spectrum (400–700 nm) to enhance overall photosynthetic efficiency. This improvement in photosynthesis can be attributed to the balancing of excitation between PSI and PSII, which operate in series to carry out the photosynthetic reactions (Zhen & Bugbee, 2020b). At shorter wavelengths, PSII is preferentially excited over PSI, while photons in the FR spectrum preferentially excite PSI (Zhen & Bugbee, 2020b). Imbalances in the excitation of these two systems leads to photosynthesis being rate limited by the under excited photosystem, normally PSI in the traditional definition of PAR (Zhen & Bugbee, 2020a). FR supplementation can therefore balance the excitation of PSI and PSII, leading to greater quantum yield of PSII as its reaction centers are freed up by faster re-oxidation of the plastoquinone pool (Zhen & van Iersel, 2017).

Increased photosynthetic efficiency with FR has been correlated with decreases in non-photochemical quenching (NPQ), where extra absorbed light is dissipated as heat (Gupta, 2017). NPQ is an indicator of chlorophyll fluorescence, which has been used as an indicator of photochemical efficiency (Gupta, 2017). High levels of NPQ indicate poor efficiency due to poor utilization of extra absorbed light, therefore reduction in NPQ and therefore reduction in chlorophyll fluorescence is a positive sign that photosynthetic pathways are being utilized more efficiently (Zhen & van Iersel, 2017). It should be noted that the stimulating effects of supplemental FR are dose-dependent, and that once balance is restored between PSI and PSII, additional FR has no further benefits (Zhen & Bugbee, 2020a).

While the inclusion of FR in spectral recipes has distinct advantages, there are some trade-offs (Vincenzi et al., 2024). Increases in leaf area can be advantageous for radiation capture, but longer stems and larger, yet thinner leaves are structurally weaker (Zhen & Bugbee, 2020b). Plants exposed to low R: FR ratios are more susceptible to disease and infection as well, especially in dense environments where shade avoidance is more common (Demotes-Mainard et al., 2016). The defense response may be undesirable by growers, as it could lead to more

drought-prone crops and less yield per harvest (Casal, 2013; Demotes-Mainard et al., 2016). It should be noted that excessive FR can potentially overexcite PSI over PSII, leading to the same rate-limiting of photosynthesis when PSII was overexcited compared to PSI (Zhen & van Iersel, 2017).

Despite the potential deleterious effects of FR, the encouragement of certain shade avoidance responses can be beneficial towards overall yield in a CEA facility (Lee et al., 2016; Meng et al., 2019). Commercial grow lights have begun offering FR as part of their spectral recipes to induce some shade responses for leafy green production (Kelly & Runkle, 2024). FR in low doses can induce leaf expansion and greater resource allocation away from roots, leading to plants with large leaves at large mass (Demotes-Mainard et al., 2016). Growers can mitigate the downsides of shade avoidance syndrome through strict pest management procedures and plant quality control. Spectral recipes can be optimized to isolate the beneficial effects of FR while mitigating any damaging effects through supplementation of other wavelengths.

2.4.2 Light Intensity

In combination with spectral distribution, light intensity is one of the major components in CEA lighting (Sago, 2016). This parameter is normally measured as PPFD, or the flux of photosynthetically active photons per unit area per unit time ($\mu\text{mol m}^{-2} \text{s}^{-1}$) (Sager & McFarlane, 1997). While a given lighting spectrum can contain the necessary photons that induce photosynthesis, growers need to consider how much of it to supply plants to maximize biological processes like photosynthesis without damaging crops or wasting energy (Sago, 2016). This is particularly true as plants can only utilize so much light for photosynthesis (Kozai, 2016). At low intensity levels starting from the light compensation point (LCP) where photosynthetic rate is zero, plants are able to make maximal gains in photosynthetic rates (Kozai et al., 2016). These gains rapidly approach some limit asymptotically once intensity reaches a maximum point, often referred to as the light saturation point (LSP) as shown in Figure 2.8 (Matsuda, 2016). Therefore, it is in the best interest of growers to choose a light intensity to not exceed the LSP.

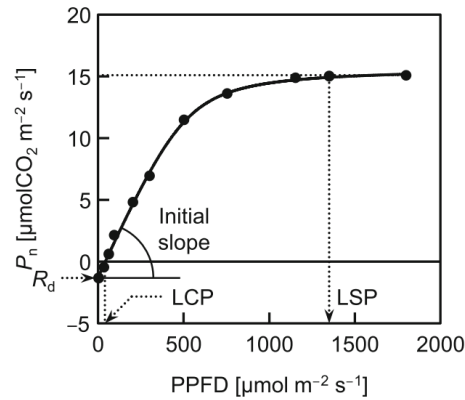


Figure 2.8. Net photosynthetic rate vs photosynthetic photon flux density (PPFD) for spinach, with light compensation point and light saturation point labeled. (Matsuda, 2016)

The effects of varying intensities have been well studied for leafy greens in different environments and CEA facilities. Universally, increasing light intensity is associated with improvements to plant growth and performance across most species of plants (Miao et al., 2023; Zhou et al., 2019). Increasing light intensity from 100 to 300 $\mu\text{mol m}^{-2} \text{s}^{-1}$ increases transpiration rates of lettuce in a plant factory setting (Miao et al., 2023; Park & Lee, 1999). At higher light intensities, without any additional optimization controls like air velocity, physiological issues such as tipburn become more prevalent as growth rate begins to outpace nutrient transport (Frantz et al., 2004). This is true for most leafy greens, and the inclusion of additional preventative measures to increase gas exchange rates allows light intensity to be driven even higher with less instances of tipburn (Frantz et al., 2004). While lettuce generally performs well from 200 to 600 $\mu\text{mol m}^{-2} \text{s}^{-1}$, studies have demonstrated that including air movement systems can allow the crop to be grown as high as 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Frantz et al., 2004).

Like many environmental parameters, both excessive and insufficient lighting leads to a different stress response in crops that reduce yield and quality. High intensities can lead to excessive rates of light absorption by photosynthetic apparatuses, particularly by PSII which is more sensitive to light (Kaiser et al., 2015). As a consequence, stress responses occur in the plant that reduce the efficiency of photosynthesis as the plant attempts to reallocate the excess energy by other means to protect PSII from further damage (Kelly & Runkle, 2024). This is

normally mediated through the previously discussed heat-release mechanism of NPQ, where high values indicate low photosynthetic efficiency (Kaiser et al., 2015). A limit appears to exist, and continual prolonged exposure to high intensities can cause continuous accumulation of energy that eventually overwhelm stress responses (Long et al., 1994). This effect is known as photoinhibition, where damage to the photosystem leads to overall reduction of net photosynthesis (Murchie & Niyogi, 2011). Consequently, plant growth and function are slowed, and combined with continual radiation exposure from high intensity lighting, desiccation and necrosis of leaf tissue is likely (Sager & McFarlane, 1997).

Low light intensity affects plants in a different way, yet with a similar end result. Instead of overabsorption of light energy from the photosynthetic apparatuses, they instead are under stimulated, as remaining under the LSP means maximal net photosynthesis is not being achieved (Kozai et al., 2016). The result of this is shade-avoidance responses, which can lead to morphological characteristics such as leaf elongation and stronger stems as plants attempt to capture more light (Sager & McFarlane, 1997). Leaf fresh mass suffers as resources are allocated towards expansion instead of structural reinforcement, and light-deficient plants become both structurally weak and susceptible to diseases and pests (Casal, 2013). While FR is a reported method for artificially inducing shade avoidance responses, the difference is that FR can be used to supplement photosynthetically optimized spectra, as opposed to insufficient lighting intensity across the board.

2.4.3 Daily Light Integral and Photoperiod

Daily light integral (DLI) is a major component of lighting that can affect plant growth and is normally defined as the amount of photosynthetically active photons that the plants receive over a 24-h period, expressed as $\text{mol m}^{-2} \text{d}^{-1}$ (Caplan, 2018). DLI is a combination of both lighting intensity and photoperiod, with the latter being the hours of lighting per day (Caplan, 2018). Lettuce has a traditional day and night period of 16/8 h respectively with an intensity of $150\text{--}250 \mu\text{mol m}^{-2} \text{s}^{-1}$ for a total DLI ranging from $6.5\text{--}9.7 \text{mol m}^{-2} \text{d}^{-1}$ (Dou & Niu, 2020). Research has shown that modifying both the intensity and photoperiod can lead to distinct improvements in plant growth if overall DLI is achieved (Caplan, 2018). These modifications can

include either changing the length of the day or night cycle, changing the day length, or even having multiple cycles within a 24-h period (Boucher et al., 2023; Hiroki et al., 2014). For instance, Hiroki et al. (2014) demonstrated that using a 20 h day with a 16/2 h photoperiod increased fresh mass by 30% when compared to 16/8 h. Alternatively, Boucher et al. (2023) increased the lighting cycle to 20–24 h per day while maintaining the same overall DLI, leading to increased performance across the board for all crops. The same study had a treatment replace the night period with an extremely low intensity at LCP ($\sim 20 \mu\text{mol m}^{-2} \text{s}^{-1}$), with similar positive effects to crop growth (Boucher et al., 2023). Kang et al. (2014) tested reduced photoperiod with increased light intensity from 200–290 $\mu\text{mol m}^{-2} \text{s}^{-1}$, observing that the highest level of 290 $\mu\text{mol m}^{-2} \text{s}^{-1}$ combined with a short photoperiod of 6/2 h led to similar positive yields when compared to more conventional 18/6 h photoperiods. Overall, modification of lighting strategies can lead to similar or better performance to their traditional counterparts, but the largest benefit comes from energy optimization as opposed to plant growth optimization, since lighting fixtures are often the most energy intensive systems in CEA facilities (Cowan et al., 2022).

2.5 Air and Leaf Temperature

Temperature is a critical factor in plant growth, as it plays an essential role in many factors of plant development (Hicklenton & Wolynetz, 1987). Photosynthesis and transpiration are directly influenced by temperature by modulating the degree of stomatal conductance, and a higher growing temperature has been linked with higher biomass accumulation at high PPFD (Hicklenton & Wolynetz, 1987; Tarr et al., 2023). Optimal day and night growing temperatures are additionally required for high yield and quality. Increasing temperature both during the day and night lead to increased growth parameters and greater biomass, yet increased crop susceptibility to tipburn due to rapid growth rates (Frantz et al., 2004). To combat this effect, decreasing night temperatures may lead to delayed onset or prevention of tipburn (Saure, 1998). Furthermore, interaction effects between day and night temperatures may not be significant, but the specific cultivar chosen may affect this assertion (Hicklenton & Wolynetz, 1987).

Temperature is not simply summarized by the ambient temperature of the growth space; it includes both the temperature of the ambient environment and leaf temperature. The interactions between these two temperatures have multiple direct and indirect effects on plant development, with different control mechanisms modulating the differences between them (Frantz et al., 2004; Hatfield & Prueger, 2015). The temperature difference between ambient conditions and at the leaf surface is most directly affected by the radiation emitted from the lighting source and the air movement system utilized in the growing system (Zhang et al., 2016). Both simulated models and real-life testing has confirmed that the differential in temperature between ambient temperature and at canopy level can differ from 1–2 °C, depending on the particular lighting source and air movement system used (Kitaya et al., 1998; Zhang et al., 2016). If the temperature within the microclimate of the canopy is too high, then plants will undergo water stress and close their stomata as a stress response regardless of the ambient environment temperature (Lee et al., 2013). Optimal temperature control additionally needs to be combined with other methods of boundary layer reduction, as it has been demonstrated lowering ambient temperature is insufficient to prevent tipburn (Lee et al., 2013).

2.6 Carbon Dioxide (CO₂)

CO₂ enrichment is a common practice in CEA to increase plant growth rates and yield. Plants assimilate CO₂ as one of the inputs from photosynthesis and convert it into organic compounds used to store energy and to build out its structure. While many environmental factors can affect the rate and efficiency in which a plant will assimilate CO₂, a sufficient amount is imperative in ambient conditions so that photosynthetic efficiency is not limited. Ambient atmospheric CO₂ levels hover around 400 ppm, nowhere close to the optimal level required for maximal photosynthetic efficiency for most plants (Kimball & Idso, 1983). It has been demonstrated in field studies that elevated CO₂ levels correlate to greater yield and growth rates for a wide variety of agricultural crops, with doubling of yields and greater water use efficiency if atmospheric CO₂ concentrations are doubled (Kimball & Idso, 1983). Kitaya et al. (2004) observed 1.2 times higher photosynthetic rates at 800 ppm than at 400 ppm in tomato seedlings. Greater CO₂ assimilation affects the structural development of plant, with shoot to

root ratio, leaf thickness, and stomatal density potentially modified at higher concentrations of CO₂ (Terashima et al., 2014).

The effectiveness of CO₂ on plant carbon fixation drops off at very high concentrations (Caplan, 2018). Caplan (2018) observed the largest yield and growth at 850 ppm, with increasing concentrations resulting in both sooner and greater instances of lettuce tipburn. Further studies agree that the upper limit of CO₂ supplementation lies around 1000 ppm, with a greater degree of plant damage and reduced gas exchange rates at higher concentrations as shown in Figure 9 (Mortensen, 1987; Park & Lee, 2001). This reduction is most notably associated with the well documented trend of reduced stomatal conductance at elevated CO₂ concentrations beyond 1000 ppm, although this reduction and its effects on plant development are crop dependent (Jones & Mansfield, 1970; Kimball & Idso, 1983; Mortensen, 1987; Park & Lee, 2001). While the precise mechanism of closure has not yet been fully elucidated, some hypotheses include the accumulation of starch from enhanced photosynthesis breaking down chlorophyll or the existence of mobile water soluble signals within the mesophyll that can induce stomatal closure at high CO₂ concentrations (Fujita et al., 2013; Mortensen, 1987; Terashima et al., 2014).

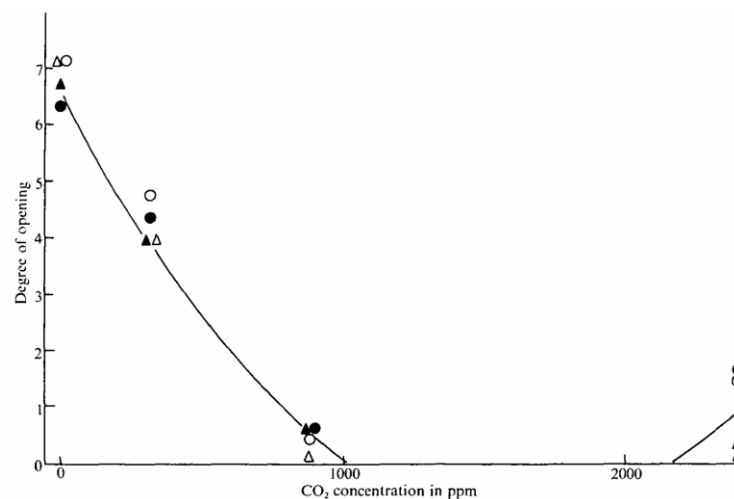


Figure 2.9. Degree of stomatal opening vs CO₂ concentration in lettuce. White and black circles represent plants grown under 330 and 1000 ppm, respectively for 3 weeks. White and black triangles represent plants grown under 330 and 1000 ppm, respectively, for 4 weeks. Points above 2000 ppm not statistically significant. (Jones & Mansfield, 1970)

Studies have explored the effects of elevated CO₂ concentrations in different controlled environment facilities (Kitaya et al., 2004; Thongbai et al., 2010). CO₂ control can be complicated by the ventilation systems used to circulate air throughout the facility. Many ventilation systems in greenhouses are designed to perform a full air exchange over set intervals for temperature and humidity control, with the inclusion of passive ventilation systems such as side vents acting as points of leakage (Thongbai et al., 2010). Therefore, CO₂ supplementation strategies in greenhouses revolve around maximizing carbon assimilation from the crops in periods of low ventilation to reduce the amount lost to the outside environment (Aikman, 1996). The uniformity of airflow system can influence the rate of CO₂ fixation by crops through reduction of the leaf boundary layer. Shibuya et al. (2006) demonstrated in a vertical farm with a VAS, airflow in an either upward or downward direction, resulted in improved dry mass and CO₂ assimilation when compared to HAS. Another study by Thongbai et al. (2010) increased both CO₂ concentration and air velocity via air circulation in a ventilated greenhouse and observed similar gains in photosynthetic performance of crops. Indoor vertical farms or plant factories generally employ HVAC systems akin to buildings that allow for a greater degree of control over ventilation (Ertle, 2023). These systems account for a much larger proportion of energy expenditures compared to greenhouses, as they need to remove excess heat and moisture from LED fixtures and transpiring plants, respectively (Graham, 2024). Consequently, similar strategies for optimal CO₂ supplementation and assimilation must still be practiced preventing unnecessary CO₂ loss and expenditures.

2.7 Conclusion

As CEA becomes more prevalent in the coming years, it is imperative that growers find ways to optimize both crop yield and quality to maintain profits. While current industry practices and academic research have demonstrated the feasibility of growing lettuce and other leafy greens in controlled environments, high density cultivation combined with rapid growth rates can lead to loss of control over vital environmental parameters. Humidity is particularly vulnerable to undesirable elevation due to the collective energy input from transpiring plants overwhelming HVAC or dehumidification systems, leading to increases in physiological disorders

such as tipburn. While other environmental parameters such as lighting and air velocity can be optimized to offset the deleterious effects of heightened humidity, new research is needed to address how these, and other environmental factors can affect the physiological and morphological development of lettuce. If plant needs are consistent with environmental factors, lettuce growth can be continually enhanced to allow for higher quality harvests with consistent environmental upkeep.

Connecting Statement to Chapter 3

The previous chapter reviewed and discussed how different environmental parameters affect plant growth in CEA, with a particular emphasis on the causes and prevention of tipburn. Underexplored technologies for enhancing plant growth including amber wavelengths were described. To fulfill the research objective stated in this thesis, the next chapter describes an investigation into the effects of novel spectra and different air velocities on lettuce growth, especially within controlled environments at elevated humidity levels. Results of this investigation will be published in a to-be determined journal for peer review.

Chapter 3 – Optimizing Lettuce Growth in Controlled High-Humidity Environments

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Abstract:

Vertical farms often face challenges with humidity management, as net transpiration rates from high-density lettuce cultivation can exceed the capacity of dehumidification and cooling systems. This study investigated the effects of different lighting treatments and air velocities on lettuce at varying levels of humidity as a means of mitigating deleterious effects of elevated humidity. Romaine lettuce (*Lactuca sativa*, cv. Teton) was hydroponically cultivated in three growth chambers under three constant relative humidities (RH): 70%, 80%, and 90%. Within each chamber, three lighting treatments were tested: white light (WL), white light supplemented with far red (WFR), and wide amber supplemented with far red (WA). Each growth tray included an air movement system constructed using a polyethylene tube attached to a duct fan to deliver two air velocity zones: low (0.5 m s^{-1}) and high (1 m s^{-1}). Under the air movement system, average air velocity measured at each plant ranged from $0.23\text{--}3.0 \text{ m s}^{-1}$. Tipburn reduction was highly effective even at the highest RH of 90%, as air movement ensured transpiration remained functional while preventing physiological disorders linked to diminished stomatal response under prolonged high humidity conditions. Results indicate that far-red supplementation to amber wavelengths may steer plant development towards leaf expansion, especially under elevated humidities. Far-red light was observed to improve CO_2 assimilation under a low Red:Blue ratio (0.15), with enhancement of stomatal conductance and comparable net assimilation to WA and WL supporting prior research on the promotive effects of blue light

on stomatal regulation as well as potential synergistic interactions with far-red. The results of this study indicate the effectiveness of air velocity levels up to 3.0 m s⁻¹ at eliminating tipburn in lettuce up to 90% RH without triggering additional stress responses. The benefits of air movement provide growers flexibility to modify lighting spectra to further optimize plant growth without concern for the negative effects of elevated humidity.

3.1 Introduction

Lettuce (*Lactuca sativa*) is a leafy green vegetable widely consumed for its crisp texture and nutritional value, containing essential minerals, vitamins, and fiber (Weber, 2016). Its suitability for high-density cultivation in hydroponic systems makes it a popular option for controlled environment agriculture (CEA) (Ahmed et al., 2020b). Many CEA facilities, notably indoor plant factories or vertical farms, struggle to maintain control over humidity and temperature due to improperly sized heating, ventilation, and cooling (HVAC) systems (Graamans et al., 2017). Environmental control is further jeopardized by high-density plant cultivation, where net energy input from transpiring plants can overwhelm the dehumidification capacity of HVAC systems leading to loss of humidity control (Graamans et al., 2017).

Transpiration is driven by the difference in vapor pressure of water between the leaf and surrounding air as defined by Monteith in Equation 3.1. The difference between χ_s , vapor pressure at the transpiring surface and χ_a , vapor pressure of the surrounding air is defined as the vapor pressure deficit (VPD), calculated in Equation 3.2 (Graamans et al., 2017; Monteith, 1965).

$$\lambda E = LAI \cdot \lambda \cdot \frac{(\chi_s - \chi_a)}{r_s + r_a} \quad (3.1)$$

$$VPD = (\chi_s - \chi_a) \quad (3.2)$$

VPD drives the movement of moisture to and from plants, with excessively low (< 0.3 kPa) or high (> 1.2–1.3 kPa) values causing dysfunction in transpiration and plant function (Grange & Hand, 1987). Ideal VPD for cultivation of different crops reportedly ranges from 0.3–1.0 kPa (Hoffman, 1979). Reducing VPD as low as 0.3 kPa in lettuce has demonstrated

improvements to stomatal conductance and photosynthetic rates while minimizing water loss (Chia & Lim, 2022; Tibbitts & Bottenberg, 1976). However, prolonged exposure to elevated humidity reduces the effectiveness of transpiration. This effect is particularly concerning in meristems enclosed by leaves or dense, tightly packed canopies that create localized hot and humid microclimate conditions (Barta & Tibbitts, 1986). Stagnant microclimates increase both the leaf boundary layer resistance (r_a) and stomatal resistance (r_s) as plants close stomata to reduce water loss, further driving down transpiration rates (Barta & Tibbitts, 1986; Fanourakis et al., 2020). Consequently, transport of xylem-mobile nutrients throughout the plant becomes less efficient (Clarkson, 1996). Calcium in particular is essential for cell wall growth in developing leaves and is transported primarily via transpiration from the roots (Clarkson, 1996). Reductions in transpiration have been correlated to lower calcium concentrations in younger leaves due to reduced transport of calcium ions, resulting in higher occurrences of tipburn, a physiological disorder characterized by wilting and necrosis in the outermost edges of the leaves (Barta & Tibbitts, 2000).

Tipburn is commonly controlled by blowing air directly onto growing lettuce to disrupt stagnant microclimate conditions, reducing the boundary layer and stomatal resistances (Ahmed et al., 2020b). Studies have reported that the required air velocity for reducing lettuce tipburn incidence lies between 0.3–1 m s⁻¹ throughout the growth cycle in a plant factory setting (Kitaya et al., 2004; Lee et al., 2013; Sohn et al., 2023). These values are normally reported as an instantaneous moment in time or as a constant state value, given that many studies utilize computational fluid dynamics (CFD) to simulate the steady-state air velocity of a given air movement system (Fang et al., 2020; Sohn et al., 2023). Changes in air velocity at the plant level over time go normally unreported, yet they may be significant for plant growth and physiological processes. A given plant can experience varying velocities from the same air movement system throughout its life cycle until maturity, such as taller plants receiving higher velocities than shorter plants, highlighting opportunities for further study.

Effects of air velocity on lettuce biomass parameters and physiological processes are variable depending on the particular design of the system. The directionality and uniformity of an air movement system can have varying impacts on plant performance, but in general any air

movement helps to alleviate tipburn occurrences (Kaufmann, 2023; Zhang et al., 2016). Higher air velocities tend to reduce physiological performance of lettuce. Peiro et al. (2020) reported an air velocity above 0.5 m s^{-1} reduced shoot dry mass, with better biomass performance around $0.3\text{-}0.4 \text{ m s}^{-1}$. Lee et al. (2013) observed reductions in fresh mass and leaf area above 1.0 m s^{-1} compared to the control at 0.08 m s^{-1} , determining a stable 0.28 m s^{-1} air speed to be ideal for tipburn reduction and biomass performance. Ahmed et al. (2020a) reported a similar positive result on biomass accumulation at an average velocity of 0.75 m s^{-1} , with a 47% reduction in leaf boundary layer resistance compared to the control. These results indicate the influence of air velocity on lettuce growth, but many of these studies test their systems under ambient humidity, normally below 70% RH. This is especially concerning as plant factories are more susceptible to loss of humidity control due to high energy requirements of dehumidification, lack of natural ventilation, and limited dimensions (Graamans et al., 2017). Higher humidities promote propagation of microbial infections, such as *Bremia lactucae* (Su et al., 2004). Additionally, elevated ambient humidity is likely exacerbated at the microclimate level, where high planting density can trap heat and moisture. Microclimate conditions are likely to worsen over time, especially since plant factories tend to lose control of humidity as plants mature due to greater biomass placing a higher energy burden on dehumidification systems (Graamans et al., 2017). This effect is outside the scope of this study, and instead we will focus on whether higher air velocities are detrimental to plant growth at higher humidity levels.

Lighting is a critical driver of plant development, playing a key role in morphological development and physiological processes (Ahmed et al., 2020b). To optimize plant productivity and yield, many CEA facilities use light emitting diodes (LEDs) as supplemental or sole-source lighting due to their relatively high electricity-to-light energy conversion factor, controllable spectra, and low surface temperature among other benefits (Kozai, 2016). Different spectra are available for LED fixtures, with red (600–700 nm) and blue (400–480 nm) wavelengths conventionally used due to their high photosynthetic efficiency (Inada, 1976; McCree, 1971). Recent studies have revealed the potential of amber (570–620 nm) wavelengths to improve plant growth, with Wu et al. (2025) reporting monochromatic amber light (half bandwidth = 10 nm) to induce the highest rates of photosynthetic efficiency within the photosynthetically active

spectrum. Reddy et al. (2024) observed similar reductions in photosynthesis at light intensities over $800 \mu\text{mol m}^{-2} \text{s}^{-1}$ for both monochromatic amber and red light, indicating similarities in photosynthetic mechanisms for both wavelengths. Additional studies reported that amber wavelengths improve biomass accumulation for tomato plants and lettuce, particularly when supplemented with blue and red wavelengths (Trumpler et al., 2024; Wu et al., 2023). These studies are relatively limited in scope in terms of spectra available during their studies, nor did they study gas exchange performance. Additional research is required to determine the full effects of amber on physiological processes and morphological development of lettuce in varying environmental conditions in CEA.

Far-red (FR), ranging beyond the visible spectrum from 700–780 nm, lies outside the range of what is traditionally considered photosynthetically active radiation (PAR); however, it functions as an important signaling mechanism for plants and as a photosynthetic enhancer (Casal, 2013; Emerson et al., 1957). Monochromatic FR alone is insufficient for plant growth, but photosynthetic performance is drastically improved when FR is combined with shorter wavelengths due to a more balanced excitation of photosystem I and II (Emerson et al., 1957; Zhen & Bugbee, 2020b; Zhen & van Iersel, 2017). FR is present in high proportion in shaded canopies, inducing shade avoidance syndrome (SAS) in plants that affect morphology through stem elongation and thinner, wider leaves to expand potential leaf area of radiation capture, and reallocation of biomass away from roots (Casal, 2013; Demotes-Mainard et al., 2016; Zhen & Bugbee, 2020b). These two characteristics of FR make it an attractive option to supplement existing LED lighting spectrums, as greater biomass can be obtained with high photosynthetic efficiency.

Characteristics of lighting spectra can have a pronounced effect on biomass partitioning within the plant. Specific leaf area (SLA) and root mass ratio (RMR) are useful metrics for characterizing the growing strategy of the plant, where SLA measures the increase in unit leaf area for a given increase in leaf mass while RMR is the ratio of dry root mass to dry stem and leaf mass (Poorter et al., 2011). SLA can be correlated linearly with leaf area and relative growth rate (RGR) across different crops (Wright & Westoby, 2001). However, SLA alone does not capture how assimilated carbon is partitioned to different parts of the plant. A model for growth

of *Arabidopsis thaliana* based on leaf area reported the relationship between leaf area and plant biomass as non-linear and varies depending on carbon partitioning (Weraduwa et al., 2015). Therefore, both SLA and RMR in combination can help characterize the nuances of how newly assimilated carbon is being distributed by the plant in response to environmental conditions.

Air velocity, amber, and far-red light (FR) are environmental parameters that have been studied individually to varying degrees, but current reported findings to date investigate ambient humidity conditions below 70% RH. While air velocity is a well-established method for reducing tipburn in leafy greens, its effects on physiological processes and final growth parameters remain variable due to differences in air movement system designs. A lack of information further exists when relating air velocity to lighting treatments, leaving potential synergistic interactions underexplored. Research on amber wavelengths has not been as extensive as other wavelengths in the PAR spectrum, and its characteristics and effects on plant growth parameters and physiological processes remain poorly understood across the board. No studies have analyzed the interactions between FR and amber, nor has tipburn been the focus of study when utilizing these wavelengths in combination with air movement systems and elevated velocities. This study aims to determine how amber and far-red wavelengths, in combination with different air velocities, influence the physiological and morphological development of lettuce under elevated humidity levels in a controlled environment.

3.2 Materials and Methods

3.2.1 Seed Germination

Lettuce seeds (*Lactuca sativa*, Teton, NORSECO, Laval, Quebec, Canada) were germinated in rockwool cubes (0.025 m × 0.025 m × 0.030 m, Grodan Inc., Kingston, Ontario, Canada) under white light at a PPFD of 200 $\mu\text{mol m}^{-2} \text{s}^{-1}$ with a 16-h photoperiod in a custom-built growth chamber (3.05 m (L) × 2.54 m (B) × 2.15 m (H)). Seeds were watered daily with a modified Hoagland nutrient solution, comprising of solution A (calcium) and B (phosphates, sulphates, and micronutrients) (Rvest Inc., Montréal, QC, Canada). Electrical conductivity (EC)

was gradually increased from 310 to 900 $\mu\text{S cm}^{-1}$ throughout the germination period (Appendix A). The growth chamber's environment was maintained at 20 ± 2 °C and 50 ± 10 % relative humidity. After 21 d, plants of similar size were randomly transplanted into all trays in each growth chamber. Experimental plants were placed in the two velocity zones (six plants per zone), and all remaining plots in the transition and edge zones were filled to prevent edge effects.

3.2.2 Cultivation schedule

Plants were watered every 3 h for 15 min during the day with the same modified Hoagland nutrient solution used for germination. EC and pH were kept at 1200 ± 100 $\mu\text{S cm}^{-1}$ and 6 ± 0.5 , respectively. Status of the nutrient solution was monitored every other day using a combined EC/pH meter (HI98129, Hanna Instruments Inc., Laval, Quebec, Canada) to ensure EC and pH were 1200 ± 100 $\mu\text{S cm}^{-1}$ and 6 ± 0.5 , respectively. Manual adjustments were made using nutrient A and B solution and 3 M nitric acid as needed. The nutrient solution for each room was replaced on day 14. Plants were monitored for tipburn development throughout their growth cycle. The temperature in the growth chamber was maintained at 20 ± 2 °C. The relative humidity and air flow rates were maintained as per the experimental treatment.

3.2.2 Experimental Setup

The experiment was conducted at McGill University's Macdonald campus (Saint-Anne-de-Bellevue, Montreal, Quebec, Canada) from June 2024 to February 2025 in three custom-designed plant growth chambers (3.05 m (L) x 2.54 m (B) x 2.15 m (H)) set at three different RH levels: 70%, 80%, and 90% RH. Each chamber was equipped with an environmental controller (Trolmaster Agro Instruments Co Ltd, Kowloon, Hong Kong, China). The 80% and 90 % RH chambers were equipped with the HCS-1 model and the 70% humidity chamber was equipped with the HCS-3 model, with only the user interface differing between models. The controllers managed the humidifiers (Lacidoll, Duluth, Georgia, USA, Levoit Corporation, Anaheim, California, USA, and Yokekon Inc, Guangdong, China), CO₂ regulator (MANATEE Inc. and Vivosun, Ontario, California, USA), and light and watering scheduling for each chamber. CO₂ tanks (Linde

Canada Inc., Saint Laurent, Quebec, Canada) were used to supply each chamber with CO₂ at 800 ppm, and temperature was controlled separately, set at 20 ± 2 °C using a split unit air conditioner (Forest Air, St-Mathieu-de-Beloeil, Quebec, Canada). Each chamber contained three growth trays attached to a recirculating hydroponic ebb-and-flow system supplied by a 60 L nutrient container (The Home Depot Inc., Georgia, USA). A styrofoam sheet was cut out to fit each tray, and 2.5" diameter holes were cut out at a planting density of 52.2 plants per meter squared (3 x 7 holes per styrofoam sheet).

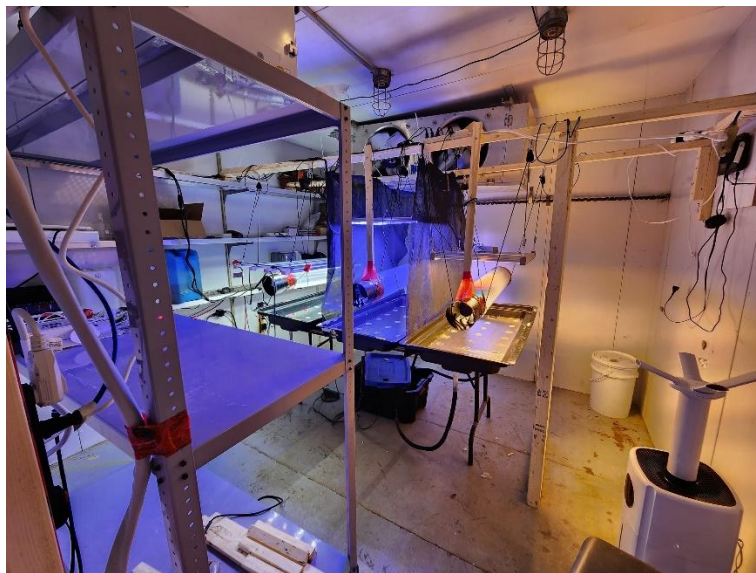


Figure 3.1. Experimental setup for 70% relative humidity (RH)

3.2.2.1 Light treatments

One of three LED treatments was used for each tray: white light (WL), wide amber supplemented with far-red (WA), and white light supplemented with far-red (WFR). WL was sourced from Futur Vert (Piedmont, Quebec, Canada) while WA and WFR LEDs were sourced from U-Technology Corp. (Calgary, Alberta, Canada). Spectral distributions and characteristics are shown in Figure 3.1 and Table 3.1, respectively. A 0.22 mm-thick black shade cloth placed between each tray prevented lighting interference. Stray light testing showed that the black shade cloth prevented 98% light interference. The intensity of each light was set at 200 $\mu\text{mol m}^{-2}$

$^2 \text{ s}^{-1}$ at tray level, and photoperiod was set at 18h/6h for a DLI of 13 mol d^{-1} . Light mapping was done before each experimental treatment to ensure uniformity of light intensity.

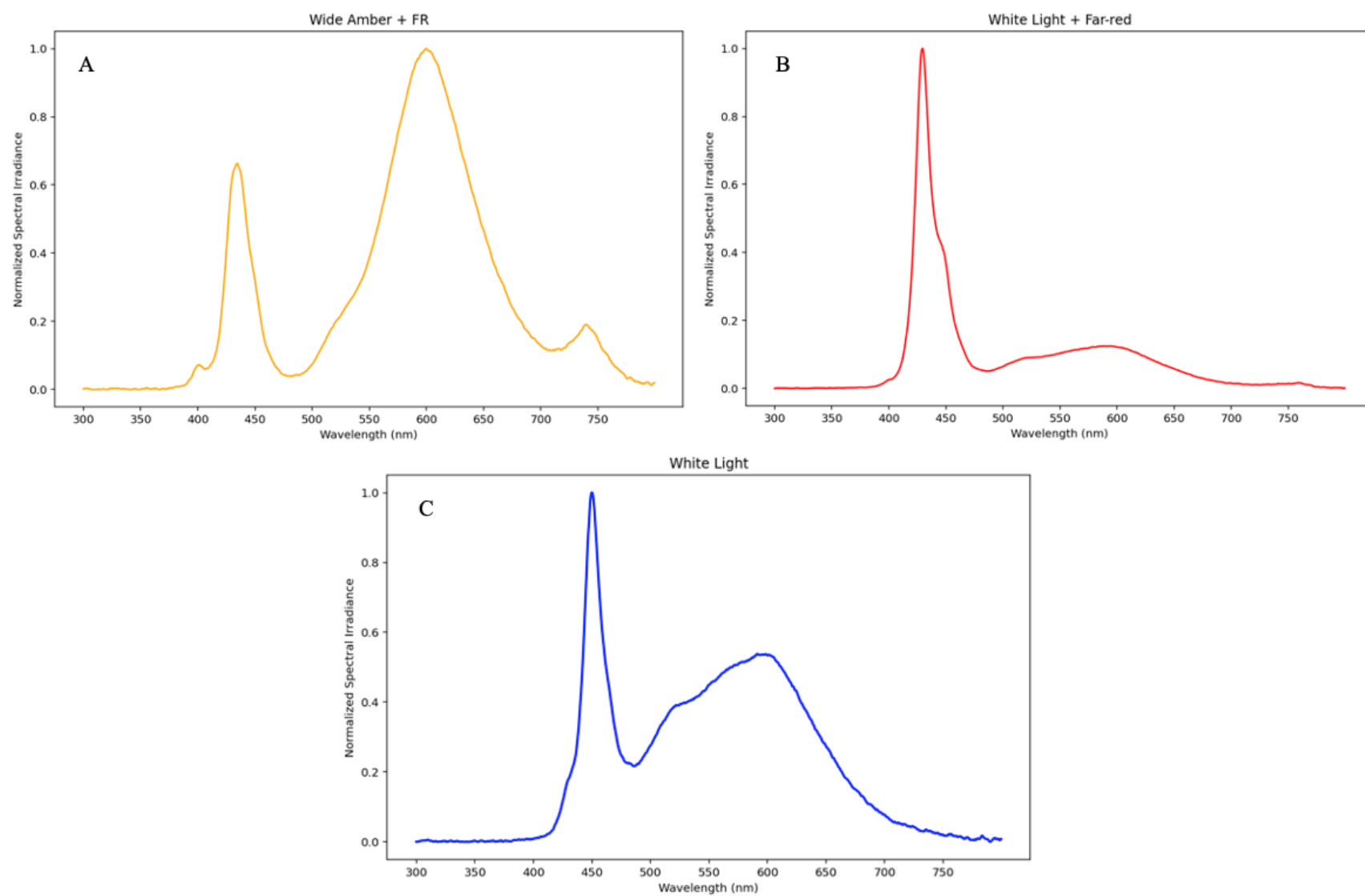


Figure 3.2. Normalized spectral irradiance of (a) wide amber and far-red (b) white and far-red (c) and white light systems taken at $200 \mu\text{mol m}^{-2} \text{s}^{-1}$.

Table 3.1. Spectral characteristics of the light systems used for the study.

LED wavelengths	Peak Wavelength (nm)	Full width at half maximum (nm)	% of PAR
Wide amber			
UV (300-399)	400.1	14.0	0.48%
Blue (400-499)	434	26.2	16.11%
Green (500-579)	-	-	21.65%
Amber (580-619)	596	88.9	28.42%
Red (620-699)	-	-	25.05%
Far-Red (700-800)	742	40.0	7.55%
Red:Blue	1.56	-	-
Red:Far-red	3.32	-	-
White + Far-Red			
UV (300-399)	399	14.03	0.67%
Blue (400-499)	429	18.79	59.51%
Green (500-579)	558	157.87	17.55%
Amber (580-619)	-	-	10.62%
Red (620-699)	-	-	8.98%
Far-Red (700-800)	761	28.1	2.30%
Red:Blue	0.15	-	-
Red:Far-red	3.91	-	-
White Light			
UV (300-399)	-	-	0.25%
Blue (400-499)	448	20.6	28.64%
Green (500-579)	-	-	31.18%
Amber (580-619)	580	156.0	19.13%
Red (620-699)	-	-	17.65%
Far-Red (700-800)	-	-	2.52%
Red:Blue	0.62	-	-
Red:Far-red	7.01	-	-

3.2.2.2 Air velocity treatments

Each tray was split into two velocity zones, high and low, using a perforated polyethylene tube ducting system constructed based on published studies (Kaufmann, 2023; Zhang et al., 2016) (Figure 3.2). A 0.1-mm thick polyethylene ducting tube (McMaster-Carr, Elmhurst, Illinois, USA) 1 m in length was constructed with three rows of 12 holes each, for a total of 36 holes per tube. Each tube was divided into two zones at the center, with hole diameters of 25.4 mm and 38.1 mm (1" and 1.5") on each side to split the cultivation area into two separate velocity zones. The center row was positioned to blow directly downward on the growth tray, while the other two rows were offset by 30 degrees to achieve better airflow uniformity. A 6-inch inline duct fan with an adjustable speed controller (HON&GUAN, Shenzhen, China) at $9.76 \text{ m}^3 \text{ min}^{-1}$ ($345 \text{ ft}^3 \text{ min}^{-1}$) was attached to one end of each tube, with the other end capped with a black 3D-printed polylactic acid cap (Bambu Lab Inc, Shenzhen, China) to ensure air exited out of the perforations. Black was chosen for the cap to prevent light reflection from the lighting treatment. Figure 3 summarizes the experimental treatments in each growth chamber and how each tray was individually configured by the velocity treatment. The transition zone was where the holes on the polyethylene tube switched from 25.4 mm to 38.1 mm (1" to 1.5"), therefore the three plants in this zone were not used as experimental units.

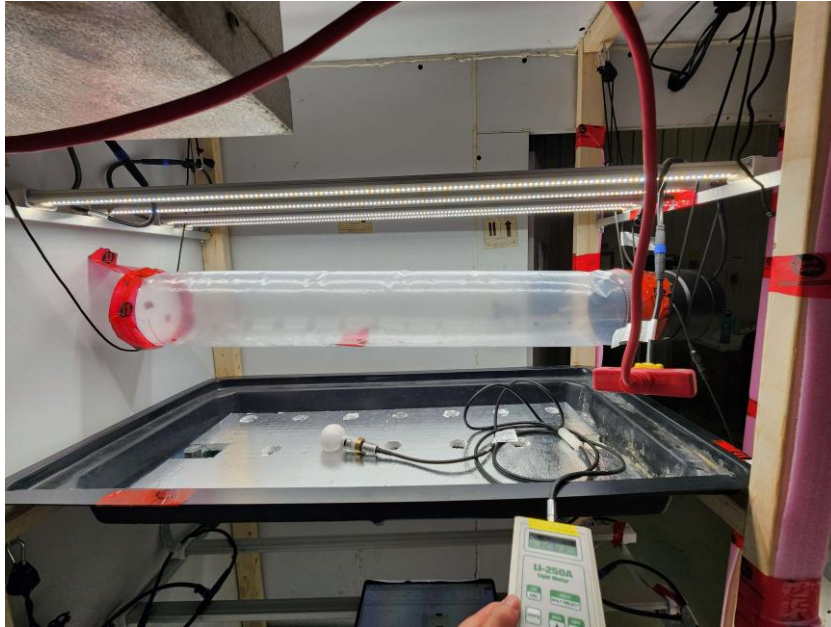


Figure 3.3. Polyethylene tube air movement system setup.

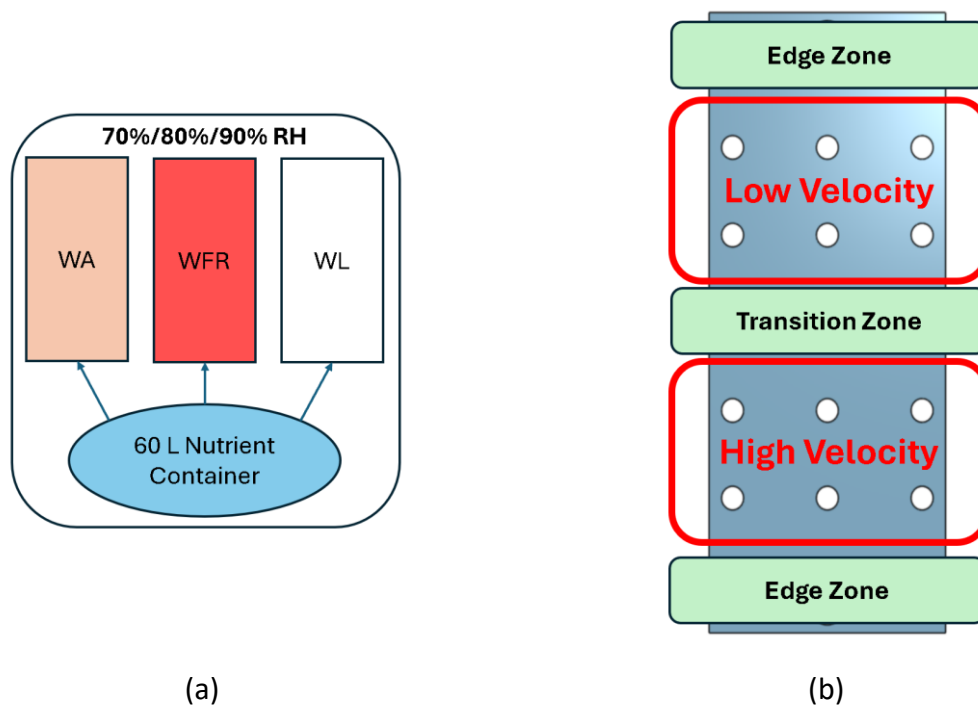


Figure 3.4. (a) Number of trays per chamber and their lighting treatment and (b) configuration of each individual tray.

3.2.3 Data Collection

3.2.3.1 Monitoring

Over a 28-d growth cycle, relative humidity, temperature, and CO₂ were tracked using the built-in data logging systems of the Hydro-X controllers. Detailed performance of environmental control in each room is presented in Appendix B. Light and air velocity mapping for each plot was conducted weekly to track changes in both light intensity and air velocity respectively. Light intensity was measured using a LI-193 spherical underwater quantum sensor connected to a LI-250A light meter (LI-COR Biosciences, Lincoln, NE, USA). Air velocity was measured using a single-directional anemometer (SD-4214, Reed Instruments, Sainte-Anne-de-Bellevue, Quebec, Canada). For air velocity, three measurements were taken per plot and averaged, and air velocity directly at the outlet of the tube was also measured to ensure no degradation in performance of the fans.

3.2.3.2 Growth Parameters

Lettuce was harvested 28 d post-transplant (49 d after seeding), and five plants from each velocity zone were used to measure the following parameters: fresh mass (leaves and roots), dry mass (leaves and roots), plant height, head diameter, leaf count, leaf area, specific leaf area, and chlorophyll SPAD content. Plant height was measured by using a ruler from the stem portion of the lettuce to the plant's apex, and head diameter was measured using a ruler. Dry mass for leaves and roots was obtained by drying the samples at 65 °C for 72 h in an isotherm oven (Fisher Scientific, Waltham, MA, US) or until a stable mass was obtained. A rockwool cube was soaked in water then dried in the oven to obtain its fresh and dry mass to approximate the root mass. Specific leaf area was obtained by dividing the leaf area by the leaf dry mass for each plant, as shown in Equation 3.3.

$$\text{Specific Leaf Area (SLA)} = \frac{\text{Leaf Area (cm}^2\text{)}}{\text{Leaf Dry Mass (g)}} \quad (3.3)$$

The number of mature leaves was counted for each plant, and six leaves that were most representative of the average leaf size of the plant were selected for leaf area analysis. Digimizer Image Analysis Software (MedCalc Software Ltd., Ostend, Belgium) was used to automate leaf area calculations. Before images were imported into Digimizer, all images were preprocessed in Photoshop (Adobe Inc., San Jose, CA, USA) to increase the contrast of the green pixels from the rest of the image. This was done to prevent shadows, debris, or other imaging artifacts from interfering with binarization. Figure 3.4 shows the process from raw image to binarized image that was used to obtain an average leaf area from the six leaves chosen.

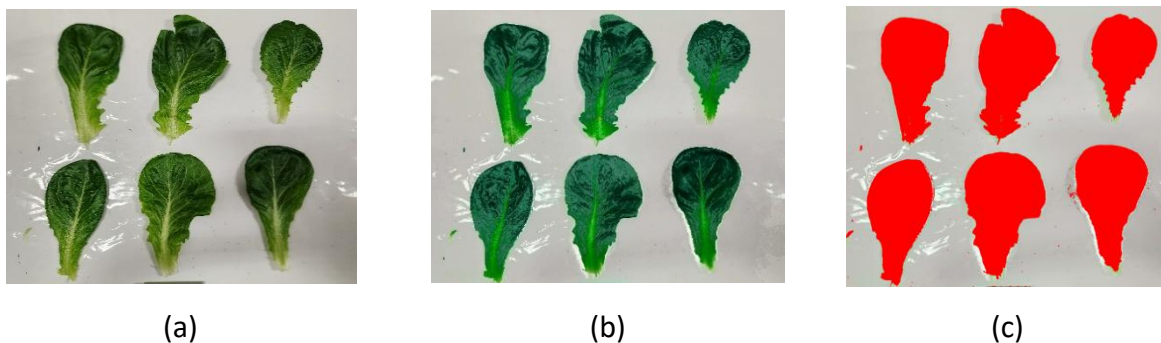


Figure 3.5. Leaf area measurement pipeline (a) raw image (b) preprocessed image in photoshop to highlight green pixels and reduce shadows (c) final, binarized image for leaf area measurement.

Chlorophyll (Chl) was measured using a soil plant analysis development (SPAD) chlorophyll meter (Konica Minolta Ltd., Saint-Laurent, QC, Canada) was used to measure the average chlorophyll (Chl) content of the six leaves from each plant used for leaf area analysis.

Pretreatment for total chlorophyll, carotenoid, carbohydrates, and anthocyanins measurement was performed the same. Samples were first stored in a -80°C freeze until ready for processing, then freeze-dried at -60°C under vacuum for 3 d (Christ Gamma 1–16 LSC, Shropshire, UK). Dried leaves were ground using a mortar and pestle. For total chlorophyll and carotenoid quantification, 0.3 g of ground powdered leaves was measured using a ENTRIS8201 scale (Sartorius Lab Instruments, Göttingen, Germany), and intubated in the freezer for 24 h at -20°C in 5 mL of anhydrous methanol. Samples were then vacuum-filtered using Whatman TM

filter paper (Thermo Fisher Scientific, Waltham, MA, US), and absorbances read at 470, 653, and 666 nm using a microplate scanning spectrophotometer (PowerWavex340, Agilent Inc, Santa Clara, CA, USA). Calculation of chlorophyll a and b and carotenoids was done using Equations 3.4, 3.5, and 3.6, respectively (Lichtenthaler & Wellburn, 1983). Results were expressed as mg per g_{dry mass}.

$$C_a = 15.65A_{666} - 7.34A_{653} \quad (3.4)$$

$$C_b = 16.37A_{653} - 11.21A_{666} \quad (3.5)$$

$$C_{caratenoids} = \frac{1000A_{470} - 2.86C_a - 129.2C_b}{245} \quad (3.6)$$

Calculation of anthocyanin concentration was done following the procedure of Teng et al. (2005). 0.3 g of ground powdered leaves were extracted for 24 h at 4 °C in 5 mL of 1% (v/v) hydrochloric acid in methanol. The extract was centrifuged at 5000 rpm for 40 m at 4 °C, then the absorbance of the supernatant was read at 530 nm and 657 nm. Equations 3.7 and 3.8 were used to calculate relative anthocyanin concentration and amount, respectively (Teng et al., 2005). Results were expressed as relative anthocyanin amount per g_{dry mass}.

$$[Anth_{relative}] = A_{530} - (0.25 * A_{657}) \quad (3.7)$$

$$Anth_{relative} = [Anth_{relative}] * V_{extract\ total} \quad (3.8)$$

3.2.3.3 Gas Exchange Measurements

Leaf photosynthesis measurements were collected immediately before harvest (Day 26-27) using the LI-6800 portable photosynthesis system (LI-COR Biosciences). One plant from each velocity zone was selected, and measurements were taken in situ using the 9 cm² leaf clip attachment on the leaf closest to the edge. The top, transparent area of the leaf chamber was kept exposed to the plant's respective light treatment. Once internal leaf chamber conditions stabilized, gas exchange flux was measured for 15 min at a sampling rate of one sample every 5 s. The leaf chamber was set at 50% humidity, temperature of 22 ± 2 °C, airflow at 800 µmol s⁻¹, and CO₂ concentration at 800 ppm. Measurements were taken throughout the day period (6:00–0:00) in both the morning (9:00–12:00) and afternoon (13:00–17:00).

3.2.4 Statistical Analysis

A completely randomized experimental design was used for this study to determine the effect of relative humidity and air velocity on the growth parameters of lettuce. The experiment was repeated three times. Statistical analysis was performed using a two-way analysis of variance (ANOVA) for humidity and lighting ($\alpha = 0.05$) using JMP software (Version 18, JMP Statistical Discovery LLC, Cary, NC, USA). For statistically significant data, Student's t-test was conducted to evaluate significant differences within treatments. Air velocity was analyzed separately using a regression analysis between the average air velocity per plot and a given morphological or physiological parameter for different combinations of humidity and lighting.

3.3 Results

This study evaluated the performance of an air movement system and its impact on tipburn occurrence and plant growth, while examining the effects of elevated humidity, lighting spectra, and air velocity on the morphological and physiological development of lettuce. Growth parameters and gas exchange performance for lettuce plants cultivated under RH of 70%, 80%, 90% and lighting treatments of wide amber supplemented with far-red (WA), white light supplemented with far-red (WFR), and white light (WL) were evaluated. Table 3.2 presents the analysis of variance (ANOVA) for the growth parameters where at least one independent term was statistically significant, while Table C1 and C2 (Appendix C) present the full ANOVA and the effect details of the independent terms, their interactions, and the quadratic term of humidity.

Table 3.2 Regression and model analyses for growth parameters where at least one independent term was statistically significant ($p < 0.05$). p-values represent the best fitting model where non-significant terms were removed ($p > 0.05$). Rows ordered by adjusted R^2 .

Growth Parameter	Effect Test (Significance)				Model Analysis		
	Lighting	Humidity	Lighting*Humidity	Humidity*Humidity	ANOVA	R^2	Adjusted R^2
Head Diameter (cm)	0.03	0.09	0.27	0.85	0.03	0.23	0.17
Leaf Area (cm ²)	0.01	0.61	0.26	0.25	0.01	0.28	0.22
Number of Leaves	0.15	< 0.01	0.75	0.64	< 0.01	0.28	0.25
Chlorophyll (SPAD)	< 0.01	0.02	0.96	0.86	< 0.01	0.47	0.40
Height (cm)	0.04	< 0.01	0.28	0.82	< 0.01	0.53	0.47

3.3.1 Fresh and Dry Mass

Fresh and dry mass are vital measures of plant performance, with high biomass accumulation occurring under optimized environmental conditions. Dry mass was not significantly influenced by any treatment based on model analyses ($p > 0.05$), and pairwise comparison of means using Student's t-test revealed no significant differences between treatments ($p > 0.05$). Fresh mass did not significantly differ between lighting treatments at any humidity, but plants under WA at 70% RH (91.97 ± 20.02 g) and 90% RH (94.69 ± 4.75 g) were significantly higher than WFR at 80% RH (65.91 ± 12.42 g; $p < 0.05$). Model analyses show no significant effect of lighting or humidity on fresh mass or dry mass ($p > 0.05$).

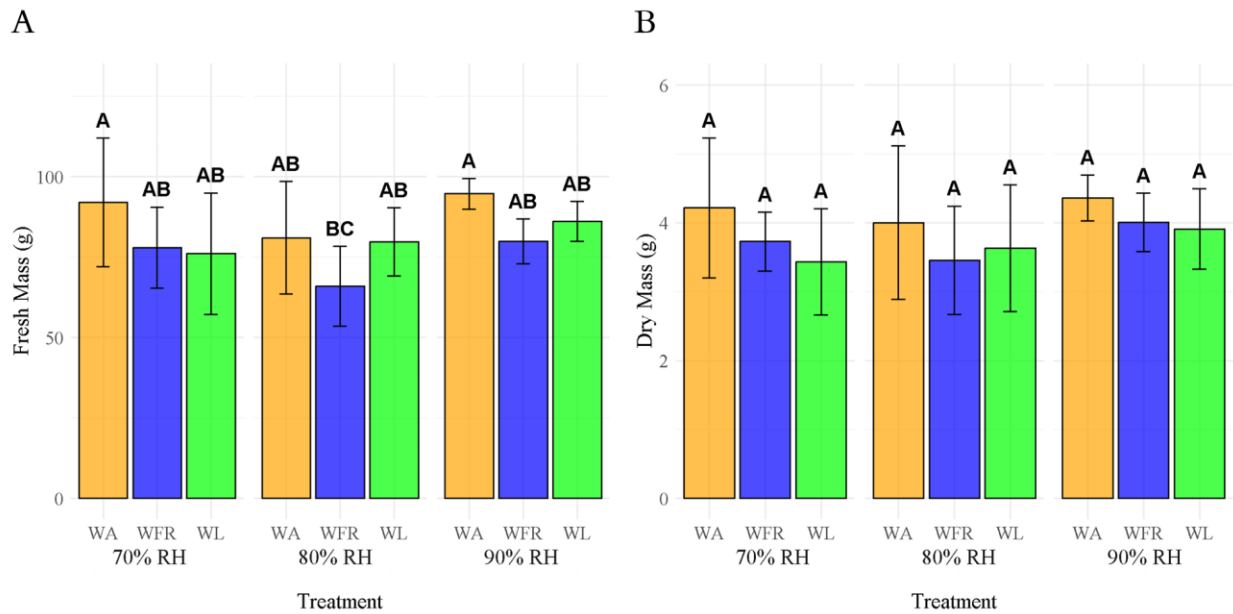


Figure 3.6. Effects of humidity and lighting treatments on (a) fresh mass (b) dry mass. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity.

3.3.2 Fresh and Dry Root Mass

Root mass is an important indicator of plant health by informing on its ability to uptake nutrients effectively. Pairwise comparison of means using Student's t-test shows a significant difference between fresh root mass under WFR at 80% RH (24.41 ± 1.82 g) and at 90% RH (29.02 ± 1.86 g; $p < 0.05$). Fresh root mass under WA at 80% RH (23.84 ± 5.49 g) was significantly lower than under WFR at 90% RH ($p < 0.05$). Increasing humidity while keeping the lighting treatment fixed did not lead to significant increases of fresh root mass accumulation ($p > 0.05$). Plants under WA at 90% RH (1.02 ± 0.33 g) had significantly lower dry root mass than WFR at 90% RH (1.58 ± 0.37 g; $p < 0.05$), WA at 70% RH (1.63 ± 0.30 ; $p < 0.05$), and WL at 80% RH (1.63 ± 0.22 g; $p < 0.05$). Otherwise, dry root mass was not strongly influenced by humidity or lighting. Model analyses showed that lighting and humidity did not significantly influence fresh or dry root mass. Interaction effects and quadratic effects of humidity were not significantly impactful on either parameter ($p > 0.05$).

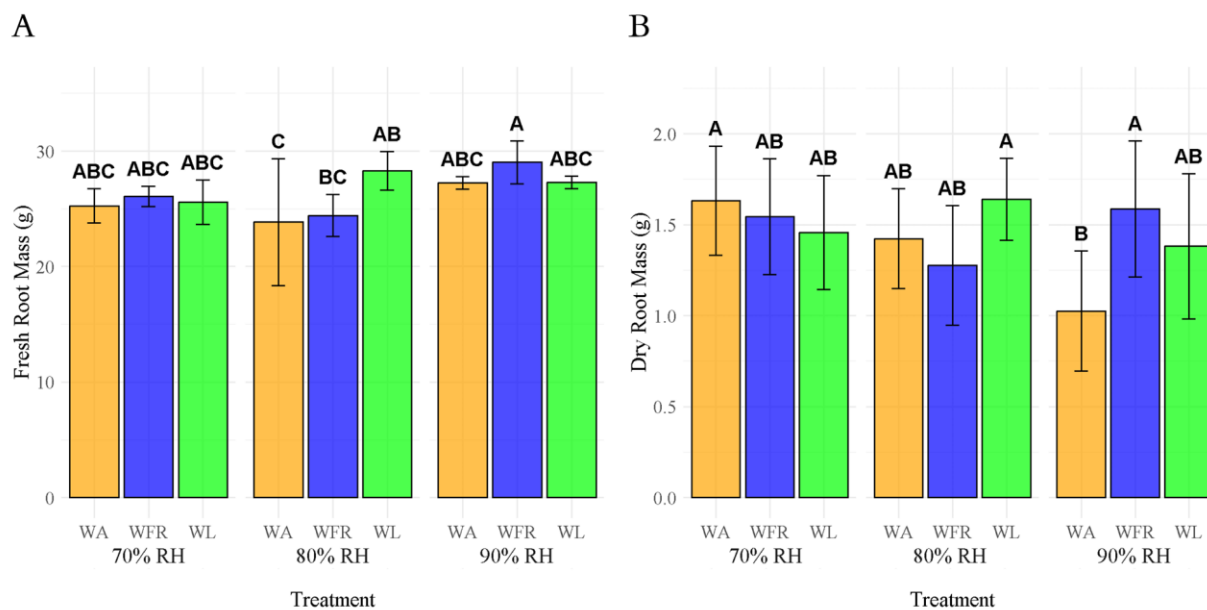


Figure 3.7. Effects of humidity and lighting treatments on (a) fresh root mass (b) dry root mass. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity.

3.3.3 Leaf Area, Number of Leaves, and Specific Leaf Area (SLA)

Morphological characteristics of leaves, such as leaf area or specific leaf area (SLA), are indicative of plant growth strategies depending on how conducive environmental conditions are for growth. Low lighting conditions may promote high leaf area, while larger SLA may be correlated with a high relative growth rate. WA generally promoted a higher number of leaves at higher humidities than WFR and WL. This is despite model analyses indicating only humidity has a significant effect on the number of leaves ($p < 0.05$). The number of leaves under WA at 90% RH (15.63 ± 1.27) was significantly higher than all lighting treatments at 70% RH ($p < 0.05$). Plants under WFR and WL did not have a significantly different number of leaves across all humidities ($p > 0.05$).

Model analyses demonstrate leaf area was significantly affected by lighting ($p < 0.05$). Plants under WA generally had greater leaf area than WL and WFR at higher humidities. Increasing humidity had no significant effect on leaf area under the same lighting treatment ($p > 0.05$), and no significant differences were observed between lighting treatments at 80% RH. Leaf area was highest under WA at 70% RH ($123.7 \pm 16.77 \text{ cm}^2$) and was significantly higher than WFR at 80% RH ($97.0 \pm 13.62 \text{ cm}^2$; $p < 0.05$) and 90% RH ($99.58 \pm 9.25 \text{ cm}^2$; $p < 0.05$). It was 18.6% higher than WL at 70% RH ($102.7 \pm 14.98 \text{ cm}^2$; $p < 0.05$), but neither of them significantly differed at 80% or 90% RH ($p > 0.05$).

Model analyses show that no lighting or humidity treatment, their interactions, or the quadratic effect of humidity had any significant effects on SLA ($p > 0.05$).

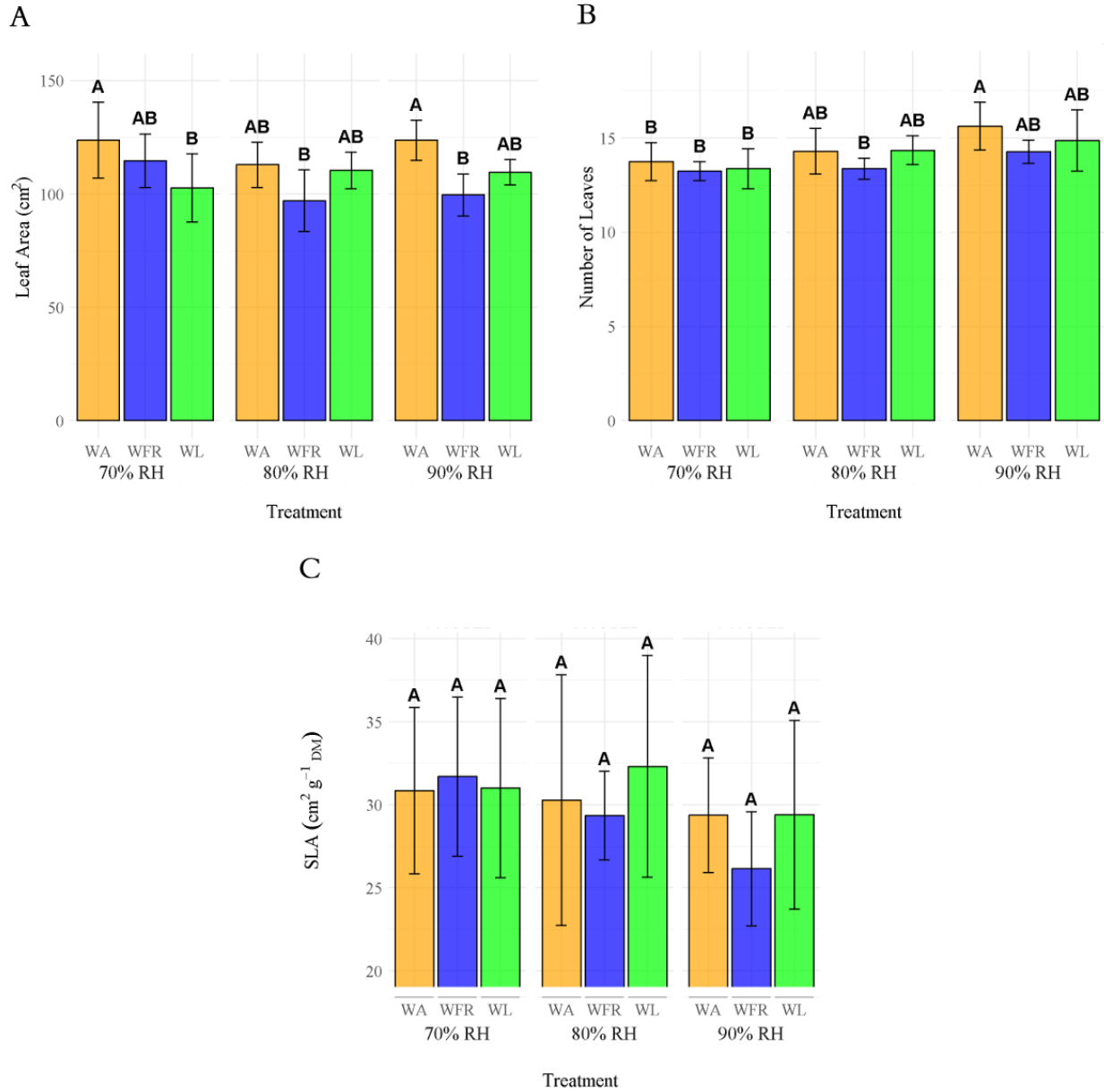


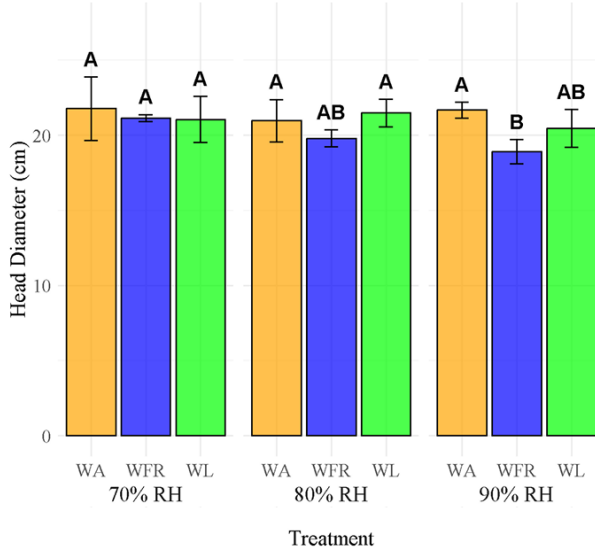
Figure 3.8. Effects of humidity and lighting treatments on (a) leaf area (b) number of leaves (c) specific leaf area. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: Relative humidity.

3.3.4 Head Diameter, Plant Height, and Root Mass Ratio (RMR)

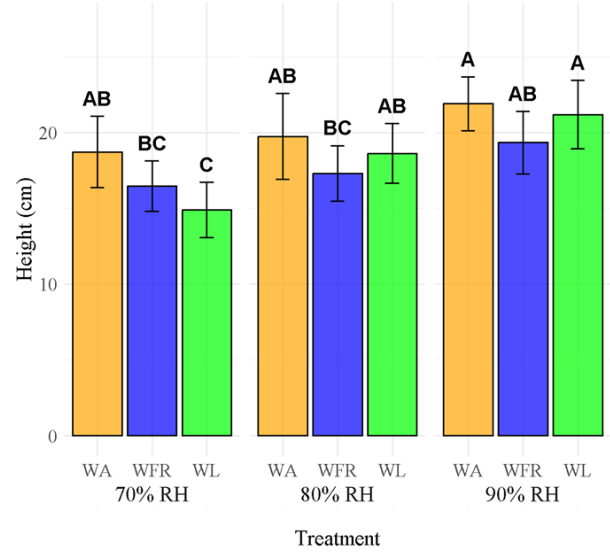
Similar to leaf parameters, plant height, head diameter, and root mass ratio give further insight on plant growth, such as how far to expand and where to allocate biomass. Model analyses reveal plant height was significantly affected by lighting and humidity ($p < 0.05$), while head diameter was only significantly affected by lighting ($p < 0.05$). The largest improvements in plant height were observed under WL. WL performed the worst under 70% RH (14.90 ± 1.81 cm) and increased significantly by 34.8% when humidity was raised to 90% RH (21.18 ± 2.25 cm; $p < 0.05$). Height of WL did not significantly differ from WA and WFR at 80% RH and 90% RH ($p > 0.05$). Head diameter was significantly affected by lighting, but distinct differences between lighting treatments were only evident at 90% RH. WFR (18.89 ± 0.81 cm) was significantly lower by 13.1% than WA at 90% RH ($p < 0.05$), while WA and WL did not significantly differ at any humidity ($p > 0.05$).

Pairwise comparison of means using Student's t-test for RMR revealed no significant differences between lighting treatments at any humidity ($p > 0.05$). Plants under WA at 90% RH had a significantly lower RMR (0.26 ± 0.11) than plants under WA (0.45 ± 0.12 ; $p < 0.05$) and WL (0.47 ± 0.10 ; $p < 0.05$) at 70% RH and WL at 80% RH (0.49 ± 0.14 ; $p < 0.05$). No treatment nor treatment combination had a significant influence on RMR based on model analyses ($p > 0.05$).

A



B



C

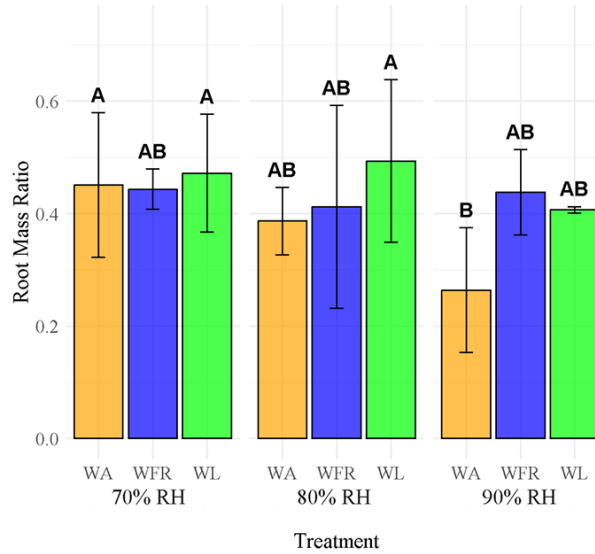


Figure 3.9. Effects of humidity and lighting treatments on (a) head diameter (b) height (c) root mass ratio. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: Relative humidity.

3.3.5 Chlorophyll

Chlorophyll (Chl) content gives insight into a given leaf's overall ability to effectively perform photosynthesis. Chl data was collected using both a SPADmeter and through plant extractions. Model analyses showed that both lighting and humidity significantly influenced Chl (SPAD) content ($p < 0.05$). Conversely, Chl a and Chl b were not significantly affected by any treatment, with no significant differences between means based on Student's t-test.

Increasing humidity under the same lighting treatment did not lead to significant differences in Chl (SPAD) ($p > 0.05$) despite humidity being significant in the ANOVA model ($p < 0.05$). Under the same humidity level, there were no significant differences in Chl (SPAD) between lighting treatments at 80% RH and 90% RH ($p < 0.05$). Plants under WL (45.99 ± 2.19) had 12.84% greater Chl (SPAD) than WA (40.44 ± 3.04) at 70% RH ($p < 0.05$). WA had significantly lower Chl (SPAD) under various treatment combinations. WA under 90% RH (38.13 ± 3.65) had significantly less Chl (SPAD) than WFR (44.28 ± 2.70) and WL at 70% RH ($p < 0.05$) and WL at 80% RH (43.65 ± 3.61 ; $p < 0.05$).

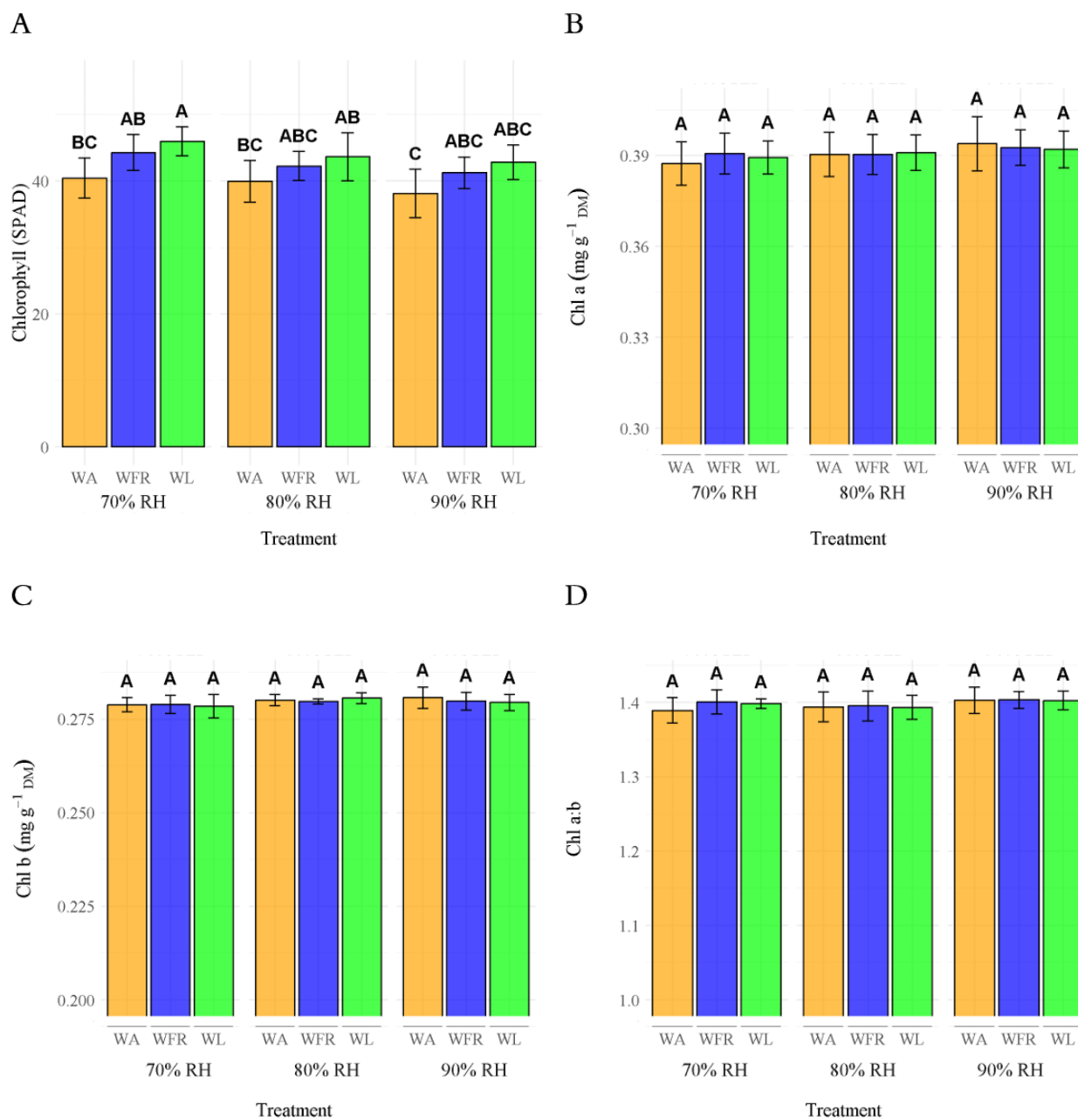


Figure 3.10. Effects of humidity and lighting treatments on (a) chlorophyll (SPAD) (b) Chl a (c) Chl b and (d) Chl a:b. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity.

3.3.6 Anthocyanin and Carotenoids

Carotenoids are vital for photoprotection of the photosynthetic apparatus by quenching excess light energy, while anthocyanins act to attenuate incoming light in high-light environments. Pairwise comparison of means showed no significant differences between treatments for any either parameter. Model analyses show that no lighting or humidity treatment, their interactions, or the quadratic effect of RH had any significant effects on carotenoid or anthocyanin accumulation ($p > 0.05$).

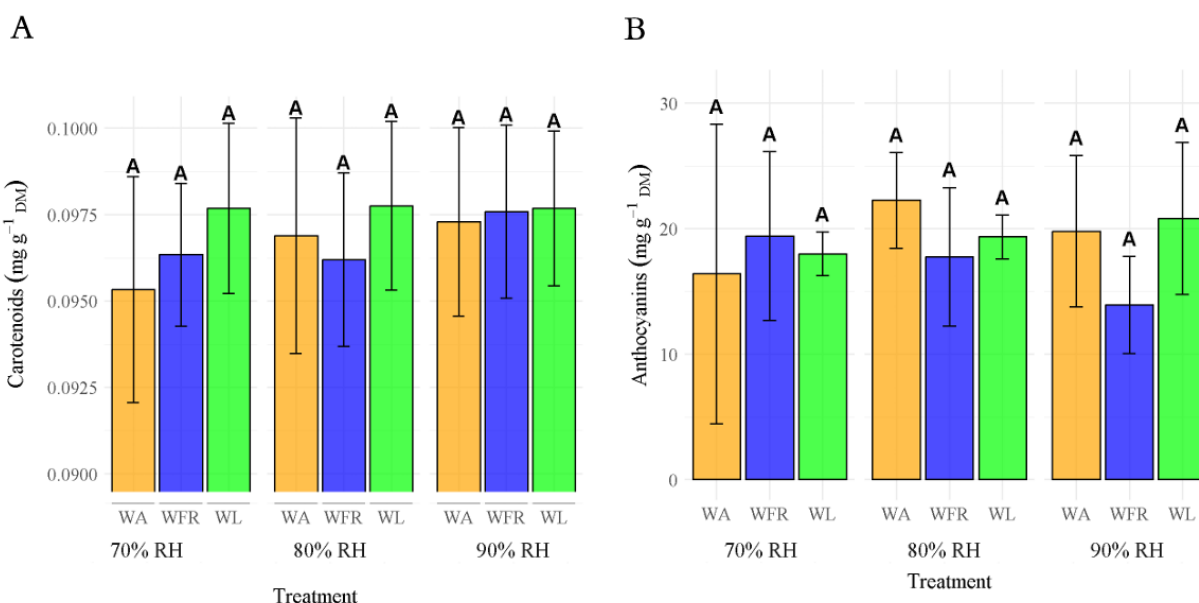


Figure 3.11. Effects of humidity and lighting treatments on (a) carotenoids and (b) anthocyanins. Significant differences in treatments denoted by different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity.

3.3.7 Assimilation Rate, Transpiration Rate, and Stomatal Conductance

Gas exchange parameters directly measure the physiological response of plants to their surrounding environment. Pairwise comparison of means showed no significant differences between treatments for any gas exchange parameter. Model analyses show that no lighting or

humidity treatment, their interactions, or the quadratic effect of RH had any significant effects on net assimilation, transpiration, or stomatal conductance ($p > 0.05$).

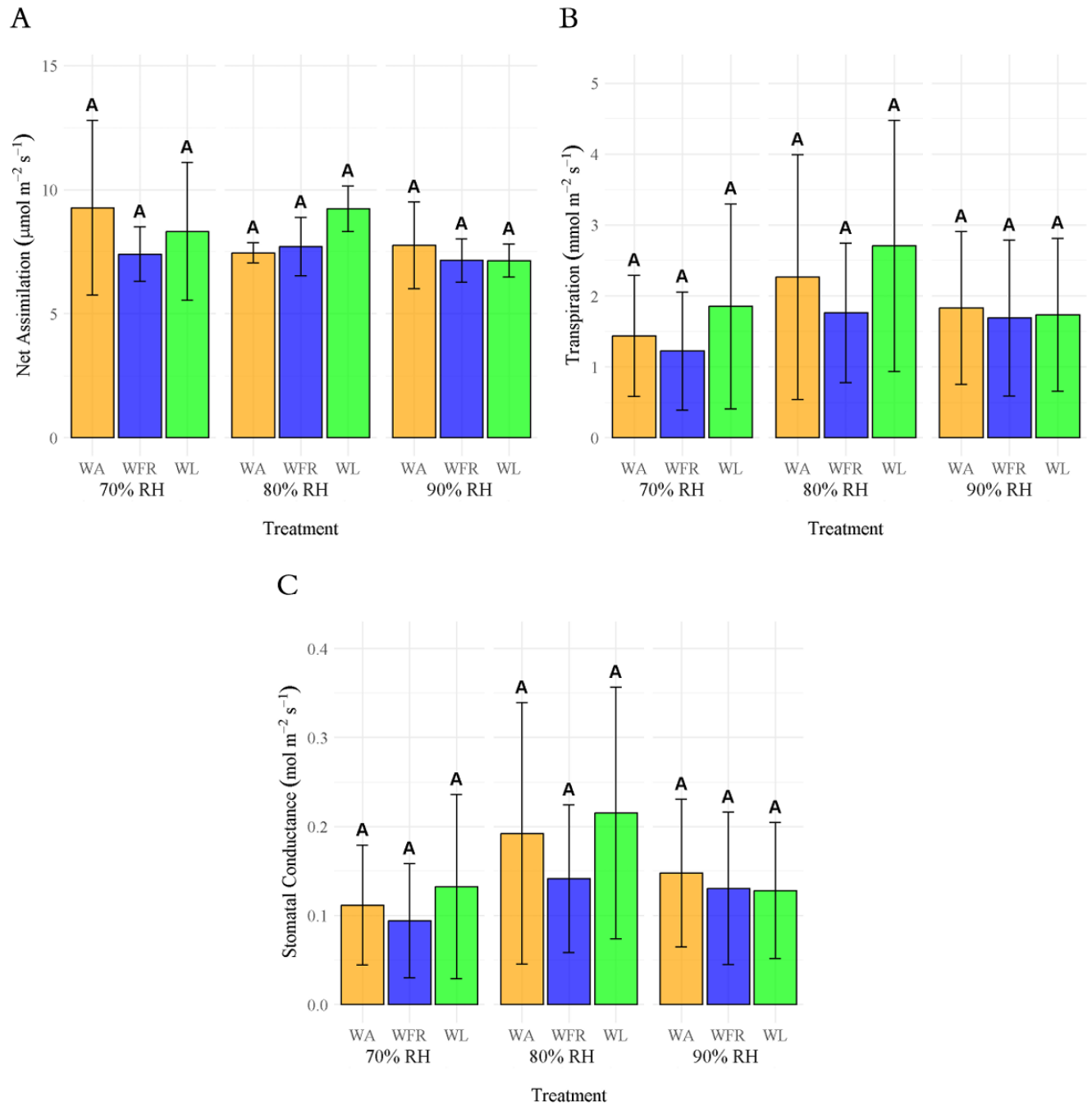


Figure 3.12. Effects of humidity and lighting treatments on (a) net CO_2 assimilation (b) transpiration and (c) stomatal conductance. Significant differences in treatments denoted by

different letters using Student's t-test. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity

3.3.8 Tipburn and Air Velocity

Table 3.3 reports the average mean air velocity and standard deviation throughout the whole trial from all replicates for each treatment combination and velocity zone. Figure 3.12 shows how average air velocity at plant level changed in one tray (Appendix D for more information).

Table 3.3 Overall air velocity statistics across all replicates (m s^{-1}). WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity

Velocity		70% RH		80% RH		90% RH	
		Average	CV	Average	CV	Average	CV
WA	High	1.31 ± 0.27	0.20	1.48 ± 0.36	0.24	1.30 ± 0.37	0.29
	Low	0.63 ± 0.07	0.12	0.74 ± 0.16	0.22	0.60 ± 0.16	0.27
WFR	High	1.13 ± 0.13	0.12	1.32 ± 0.39	0.29	1.53 ± 0.47	0.31
	Low	0.57 ± 0.05	0.09	0.65 ± 0.15	0.23	0.68 ± 0.15	0.22
WL	High	1.34 ± 0.18	0.13	1.50 ± 0.23	0.16	1.38 ± 0.35	0.25
	Low	0.62 ± 0.15	0.24	0.76 ± 0.19	0.25	0.63 ± 0.06	0.09

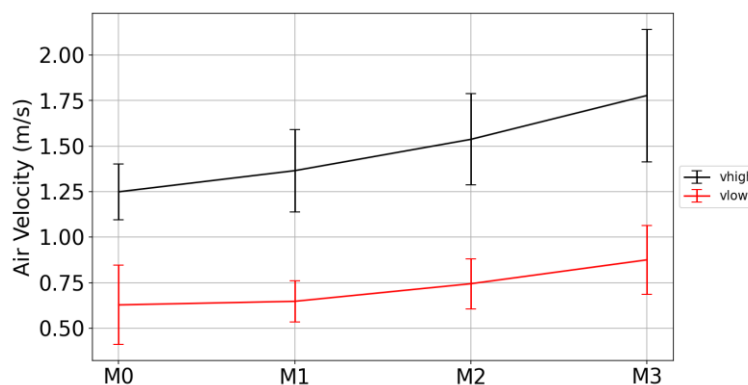


Figure 3.13. Average air velocity at plant level for 80% relative humidity (RH) and wide amber supplemented with far-red. M0, M1, etc denotes the week the measurement was taken.

Table 3.4. Average percent of tipburned plants per tray (6 plants per velocity zone). WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity

	Velocity	70% RH	80% RH	90% RH
Tray 1 - WA	High	4.17%	6.67%	25.00%
	Low	25.00%	16.67%	25.00%
Tray 2 - WFR	High	0.00%	6.67%	16.67%
	Low	0.00%	16.67%	16.67%
Tray 3 - WL	High	0.00%	0.00%	0.00%
	Low	0.00%	0.00%	8.33%

Table 3.5. Average air velocity per treatment combination overlayed over frequency of tipburn occurrences. WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity

		Velocity		
WA	High	1.31 ± 0.27	1.48 ± 0.36	1.30 ± 0.37
	Low	0.63 ± 0.07	0.74 ± 0.16	0.60 ± 0.16
WFR	High	1.13 ± 0.13	1.32 ± 0.39	1.53 ± 0.47
	Low	0.57 ± 0.05	0.65 ± 0.15	0.68 ± 0.15
WL	High	1.34 ± 0.18	1.50 ± 0.23	1.38 ± 0.35
	Low	0.62 ± 0.15	0.76 ± 0.19	0.63 ± 0.06
		70%	80%	90%
Relative Humidity (RH)				



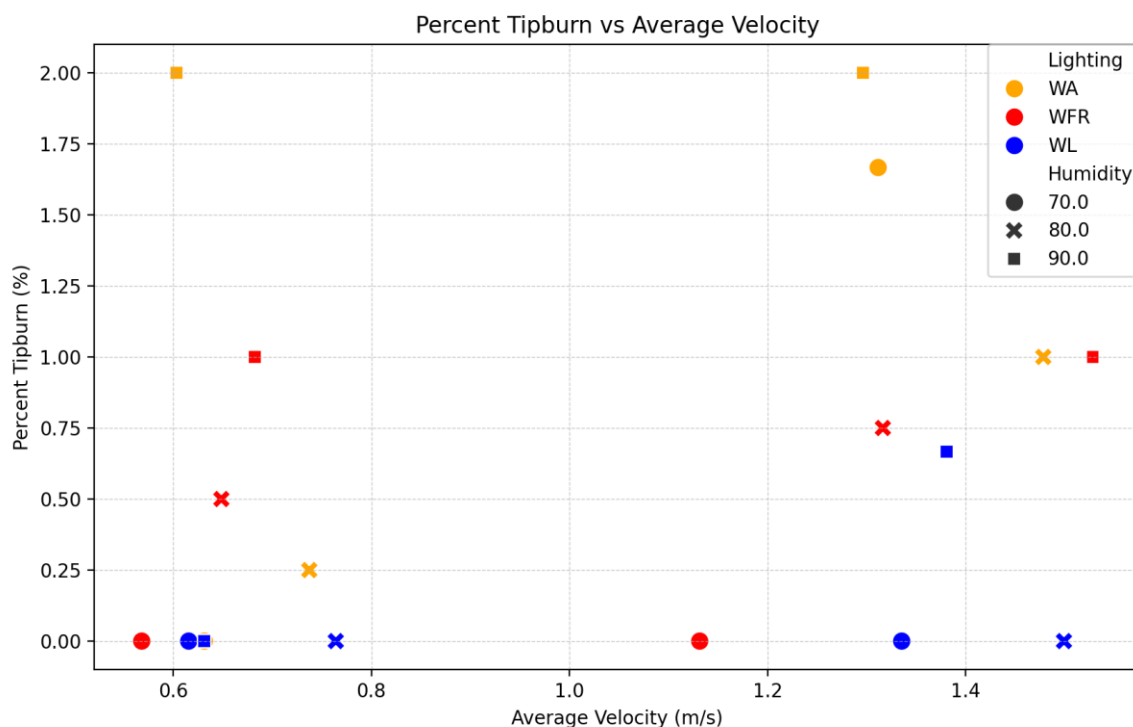


Figure 3.14 – Average air velocity per treatment combination plotted against percent tipburn

Performance of the different air movement systems varied between different growth trays, with coefficient of variation peaking at $\pm 31\%$. High and low velocity zones remained distinct from each other across all replicates, with high velocity zones at least 100–124% higher than low velocity zones. Average air velocity per plot ranged from 0–3 m s^{-1} . Regression analyses revealed no significant relationships between air velocity and any growth parameter or gas exchange performance ($p > 0.05$). Figures E1 and E2 (Appendix E) provides examples of average air velocity plotted against fresh mass grouped by humidity and lighting, further demonstrating the lack of correlation. Tipburn was effectively minimized by air velocity across all treatment combinations (Table 3.5). On average for a given trial, 9% of plants exhibited symptoms of mild tipburn with no cases of severe damage. Lighting appeared to have a greater influence on tipburn occurrence compared to humidity. WA exhibited the highest average number of tipburn occurrences at 25% of plants in a given velocity zone, followed by WFR at 16.67%, then WL at 8.33%. Elevated humidity levels increased tipburn occurrences, but plants under the high velocity treatment at 70% or 80% RH had reductions in tipburn by 10–20%.

3.4 Discussion

3.4.1 Air Velocity

Plant growth and development are affected by different environmental factors to varying degrees. Air movement is a common method used to mitigate or prevent tipburn development in leafy greens within CEA facilities. Optimization of air velocity and distribution allow for more precise control of temperature and humidity while promoting efficient circulation of CO₂ to improve plant assimilation (Kozai et al., 1980). While designs of ventilation systems vary, studies report improvements to assimilation and transpiration rate in proportion to air velocity up to 1.0 m s⁻¹ for lettuce and tomatoes (Ahmed et al., 2020a; Kitaya et al., 2004). Average air velocity per plant in this study ranged from 0–3 m s⁻¹ throughout the growth cycle, and percent increase from start to finish which ranged from 0.23–62% (Table 3). This large variation is due to the selection of a single-directional anemometer for air velocity characterization. While omnidirectional anemometers have been demonstrated to better characterize the average velocity within a given volume by measuring air velocity from three planes, single-directional anemometers allow for characterization of the air speed at specific height levels and can better capture the air movement at the meristem of a lettuce most susceptible to tipburn (Kaufmann, 2023).

Despite variations in uniformity and magnitude between growth trays, no significant relationship was observed between air velocity and any morphological characteristics under any treatment combination. These observations are in direct contrast with prior literature that report positive correlations between air velocity and specific growth parameters like fresh mass and dry mass of lettuce accompanied by increases in carbon assimilation (Ahmed et al., 2020b; Peiro et al., 2020). Reductions in plant performance have been reported at air velocities above 1.0 m s⁻¹, likely associated with defense responses via thigmomorphogenic signaling pathways that reallocate biomass towards smaller, stronger plants to withstand increased air movement (Chehab et al., 2008; Peiro et al., 2020). These defense responses were not evident based on regression analyses between average air velocity and any growth parameter, which indicated no significant differences in plant development from 0 to 3 m s⁻¹. It is possible that thigmomorphogenic responses are not strongly triggered by the particular type of mechanical

stimulation air movement system delivers, as previous studies apply direct contact to plants to induce these defense mechanisms (Graham & Wheeler, 2017). Further study will be required to evaluate the unique response to mechanical stimulation of lettuce in CEA environments, as the upper limit of 3 m s^{-1} may not have been sufficient to produce any pronounced effects.

The major morphological effect of air velocity was tipburn reduction. Despite minor improvements in tipburn occurrences at higher air velocities under WA, overall tipburn occurrence and severity were minimal to none across all treatment combinations and replicates (Table 4). Air movement has been well characterized to improve tipburn in controlled environment cultivation of leafy greens, attributed to a reduction of the leaf boundary layer and increase in stomatal conductance to allow for greater transport of calcium from the roots to the outer leaves (Ahmed et al., 2022; Frantz et al., 2004). While transpiration and stomatal conductance were unaffected by air velocity, a possible explanation is that air movement was sufficient to eliminate stomatal and boundary layer resistances as a limiting factor to transpiration. The minimum threshold where these resistances become negligible will vary depending on the growing system and specific plant, but studies on lettuce cultivation in plant factories have reported stomatal conductance ranging from $0.03\text{--}18.50 \text{ mol m}^{-2} \text{ s}^{-1}$ without suffering significant tipburn or other physiological disorders (Ahmed et al., 2022; Kang et al., 2014). Further research is required to dial in this threshold, but for practical purposes the mere presence of air movement is sufficient to prevent most, if not all instances of tipburn for lettuce cultivation in CEA (Ahmed et al., 2020b).

3.4.2 Humidity

Humidity is a vital aspect for plant health, as ambient humidity is what determines the flow of moisture between the leaf and the surrounding atmosphere. A higher ambient humidity, i.e. a lower VPD can lead to decreased rates of transpiration and therefore higher instances of tipburn due to insufficient calcium transport (Frantz et al., 2004). High temperature and humidity microclimates surrounding plants can exacerbate reductions in transpiration by increasing stomatal closures to prevent water loss or by general malfunctions in stomatal opening (Arve et al., 2015). CEA facilities like plant factories or vertical farms are especially

susceptible to elevated humidity, as dehumidification systems often cannot keep up with the net energy input from high-density transpiring plants (Graamans et al., 2017).

In this study, despite maintaining a constant humidity level throughout the day and night cycle, no major reductions in plant morphology or physiology were observed under any humidity treatment. Tipburn was well controlled, demonstrating the effectiveness of the air movement system in preventing stagnant microclimate conditions from forming up to 90% RH. This observation is particularly relevant given that prolonged exposure to high humidity (>85%) is known to attenuate all aspects of stomatal function (Fanourakis et al., 2020). Above this threshold, studies generally report larger pore openings and bigger stomata (Fordham et al., 2001; Nejad & Van Meeteren, 2005). These changes in stomatal development are normally accompanied by reductions in the ability to close stomata that can potentially lead to excessive moisture loss (Fordham et al., 2001; Nejad & Van Meeteren, 2005). However, stomatal conductance and transpiration were not significantly affected by humidity, indicating normal physiological function even at elevated RH. These results reinforce the prior conclusion that air movement plays a critical role in regulating stomatal response and maintaining normal physiological behavior above 80% RH. Additionally, air movement likely reduced deleterious effects of open stomata by breaking up stagnant microclimate conditions, preventing excessive moisture loss and regulating transpiration.

Determining an exact RH or VPD for optimal lettuce growth is challenging due to the complex interactions of environmental factors that influence how humidity affects plant performance. Existing research suggests a tenuous correlation between increasing RH on carbon assimilation, and results of this study weaken this correlation as humidity had no significant effect on assimilation. The primary benefits of optimized humidity are likely limited to reduction of physiological disorders via enhancements in transpiration rather than direct improvements to growth (Frantz et al., 2004; Gómez et al., 2019). Therefore, humidity optimization should be framed as the range where plant function is not impeded by excessive moisture loss or insufficient transpiration. This range can generally be identified as a VPD of 0.2 to 1.0 kPa for a wide variety of crops, but results of this study indicate that this range can potentially be expanded if humidity control deteriorates (Grange & Hand, 1987).

Increasing humidity did not have a strong impact on any growth parameters; its primary influence limited to accentuating differences between lighting treatments. Root mass ratio was not significantly affected by humidity despite minor variations between treatments. Therefore, the effects of elevated humidity on biomass partitioning are likely negligible. Similar observations have been reported for white silver birches (Sellin et al., 2015; Sellin et al., 2019).

This study demonstrated that plant function well under 70 to 90% RH (VPD of 0.2–0.7 kPa) remained normal despite variations in performance. The air movement system did not enhance carbon assimilation nor improve growth parameters, but it was highly effective at attenuating or outright eliminating deleterious effects of elevated humidity through prevention of tipburn and improved stomatal function. Results do not necessarily indicate that elevated humidity is ideal for plant function. Rather, under conditions where humidity control is compromised, plant quality can still be maintained through constant air movement. Further optimizations can be achieved through the adjustment of lighting spectra to induce morphological characteristics in lettuce that might potentially be better suited for high humidity growth.

3.4.3 Lighting

Performance differences between lighting treatments highlight the role spectral composition has in plant development. No significant differences between lighting treatments were observed for gas exchange performance. This is despite the large differences across the different spectra utilized, especially under WFR with a much lower R:B ratio (0.15) compared to WA (1.56) and WL (0.62). Wavelengths in both red and blue regions are vital for biomass accumulation and photosynthetic function, where reductions in physiological or morphological development occur when either R or B dominate (Savvides et al., 2012). Plants grown under WFR only tended to underperform at 90% RH compared to WA, but development generally did not significantly differ to WL and WA at lower humidities. These observations are at odds with the current understandings of spectra supplemented with high proportions of B. B has been demonstrated to be vital for upregulation of stomatal opening, with plants under monochromatic B exhibiting normal stomatal function (Hogewoning et al., 2010; Izzo et al.,

2020; Savvides et al., 2012). However, improved stomatal regulation does not necessarily correlate to improved plant growth or assimilation. Similar studies on tomato and lettuce studying the effects of B report no significant improvements or correlations to biomass accumulation in response to increasing proportion of B, indicating the poor ability of monochromatic B to drive photosynthesis (Izzo et al., 2021; Wang et al., 2016; Wu et al., 2024).

Potential synergistic interactions with far-red supplementation may account for improvements in photosynthetic performance and comparable growth parameters under WFR to WA and WL. A study by Zhen and van Iersel (2017) supplemented FR to R:B spectrums for lettuce, reporting a 13.5% increase in net photosynthetic rate from 0–9% FR. FR has been demonstrated to improve stomatal opening and transpiration in sweet basil (*Ocimum basilicum*) (Park et al., 2023). The combined effects of FR on stomatal opening and assimilation may have potentially compensated for any reductions in gas exchange rates due to an overexposure of B. Further improvements in gas exchange can likely be attributed to the air movement system, which potentially enhanced stomatal conductance across all treatments by variable amounts due to variations in air velocity between cultivation trays. This variable improvement would have masked weaker spectral effects on stomatal characteristics, leading to a lack of significant difference in stomatal conductance across all lighting treatments despite the low R:B ratio under WFR. Therefore, the influence of FR combined with the regulatory effects of B on stomata could have raised photosynthetic performance such that there was no significant difference in assimilation under WFR compared to the other lighting treatments that ostensibly contain more photosynthetically active wavelengths.

Current studies that utilize amber wavelengths for plant growth report enhancement of biomass accumulation in lettuce and tomato plants. Higher fresh mass and leaf area are generally observed under amber wavelengths with peaks at 595 nm, with further improvements in assimilation rates upon B or R supplementation (Trumpler et al., 2024; Wu et al., 2021; Wu et al., 2024). In this study, WA and WL had peaks in amber at 596 nm and 580 nm, respectively, yet neither spectrum promoted significantly higher biomass accumulation compared to WFR. Difference between the two spectra at 70% RH were generally more pronounced, as increasing humidity eliminated most significant differences between all spectra. Of particular note are

differences in leaf area, Chl content, and height, which provided potential insight into the specific spectral differences of WA and WL and their corresponding effects on plant development.

Growth parameter performance under WA and WL were generally similar, yet their similarity cannot be explained by amber wavelengths alone. Amber photons (580–619 nm) as a percentage of the PAR spectrum were 28.42% and 19.13% for WA and WL, respectively. Research reporting on specific spectral characteristics of amber are limited; however, a study by Wu et al. (2025) identified green and amber regions at 575 nm and 595 nm, respectively, as having the highest quantum yield compared to the rest of the PAR spectrum. Green wavelengths have been characterized to have high penetration into dense canopies, making them abundant in shaded regions where they can induce shade-avoidance response like leaf elongation and enhance photosynthesis (Kim et al., 2004; Klein, 2008). Similarly, amber wavelengths have been observed to drive leaf elongation in growth trials for tomato and lettuce plants, suggesting similar absorption and signaling characteristics between amber and green wavelengths (Wu et al., 2024). The combined percentage of amber and green wavelengths for WA and WL lie at 49% and 50%, respectively, which likely accounts for the similar physiological response under these two lighting conditions.

WA included far-red supplementation, which shares key absorption and signaling characteristics with green light. Like green, far-red is efficiently transmitted and reflected by foliage due to low absorption above 700 nm, enabling it to penetrate dense canopies and accumulate in shaded regions (Demotes-Mainard et al., 2016). This spectral quality results in low R:FR ratios, which induce shade-avoidance signaling pathways that promote leaf expansion and stem elongation (Demotes-Mainard et al., 2016; Heraut-Bron et al., 1999). Moreover, low intensities of G at peaks from 520-526 nm have been demonstrated to augment far-red induced shade avoidance responses in *Arabidopsis thaliana*, lettuce, and kale (Kelly & Runkle, 2024; Meng et al., 2019). These parallels in function suggest that amber may exhibit synergistic effects with FR light similar to those observed with G.

WA in this study had the lowest R:FR ratio at 3.32 compared to 3.91 and 7.01 in WFR and WL, respectively. The differences in leaf area were more pronounced at 70% RH, where WA had

18.6% higher leaf area than WL. At 80% and 90% RH, WA and WL no longer significantly differed, and WFR performed similar to both. These results suggest potential limits of FR under the R:FR ratio used in this study, with amber or green wavelengths possibly exerting a dominating influence on leaf elongation over FR. Conversely, effects of FR may be more pronounced at higher humidity levels. WA had 12.8% more leaves at 90% RH than 70% RH, whereas the same comparison under WFR and WL did not significantly differ. This increase, combined with a lack of significant effects of lighting on SLA, hints that FR wavelengths may steer carbon partitioning towards leaf expansion under heightened humidity. This portioning hypothesis is supported by significantly lower RMR and dry root mass under WA at 90% RH compared to WL and WFR at similar or lower humidity levels. However, these reductions were not particularly pronounced, likely due to the air movement system dampening spectral responses. Combined with a lack of significant effects of lighting on RMR implies that air movement reduced the plant's need to invest in leaf expansion when assimilation and stomatal regulation remain optimal.

FR supplementation had the most distinct effect on Chl distribution within the leaf at 70% RH. Although leaf area was greater under WA than WL at 70% RH, Chl was 12.8% lower under WA than WL. This depressive effect of WA only held at 70% RH, supporting the previous conclusion that effects of FR are limited. Prior studies report reductions in Chl concentration in various species, both overall and per unit area, upon far-red supplementation (Demotes-Mainard et al., 2016; Zhen & Bugbee, 2020b). Effects of amber on Chl content are less clear. One study by Wu *et al* reported reductions in chlorophyll content past 800 $\mu\text{mol m}^{-2}$ of wide amber light in tomato plants, while another by Trumpler *et al* noted higher chlorophyll accumulation in lettuce plants under narrow amber supplemented with blue (Trumpler et al., 2024; Wu et al., 2023). Both studies reported total chlorophyll content or chlorophyll content per leaf area based on SPAD readings, the accuracy of which can vary significantly due to environmental factors, crop-specific characteristics, and user bias (Xiong et al., 2015). Results of total chlorophyll accumulation showed no significant impact of any lighting or humidity treatment, indicating that the only major effect of amber or FR on chlorophyll was limited to its distribution throughout the leaf as opposed to total accumulation. This effect would address the lower SPAD

readings under WA, as a greater leaf area with the same amount of chlorophyll would lead to a lower chlorophyll content on a per area basis.

Total carotenoid and anthocyanin accumulation were similarly unaffected by lighting treatments. Carotenoids form integral parts of the photosynthetic apparatus, involved in both organization of the reaction centers and functioning of the light harvesting complex (Maslova et al., 2021). They also act photoprotective compounds, quenching excited chlorophyll states to reduce buildup of reactive oxygen species (ROS) and helping regulate the non-photochemical quenching of fluorescence (Maslova et al., 2021). Anthocyanins serve a similar role, acting as lighting attenuators or ROS scavengers to protect plants in high-light environments (Landi et al., 2015). G has been demonstrated to promote carotenoid accumulation of lettuce (Salehinia et al., 2024), while UV-A and low R:B have been reported to induce anthocyanin production in lettuce (Chen et al., 2019; Wang & Folta, 2013). It is likely the total PPFD of $200 \mu\text{mol m}^{-2} \text{s}^{-1}$ was insufficient to produce any significant differences in anthocyanin or carotenoid accumulation, and spectral effects of amber were not evident. However, based on reported observations of bleaching under intensities $>1000 \mu\text{mol m}^{-2} \text{s}^{-1}$ under solely wide amber by Wu et al. (2024) or prior studies on UV defense responses (Chen et al., 2019; Di et al., 2012), increasing the overall or individual wavelength intensities to an intermediate level above $200 \mu\text{mol m}^{-2} \text{s}^{-1}$ may potentially increase anthocyanin or carotenoid production.

Further research is needed to better understand the combined effects of amber and far-red wavelengths on carbon partitioning throughout plant development, including potential trade-offs such as thinner leaves or reduced assimilation efficiency at the leaf or whole-plant level. The unique spectral responses of each lighting treatment were likely attenuated or masked by the air movement system, which may have evened out distinct differences by alleviating stomatal limitations. Beyond a certain threshold, stomatal resistance may no longer constrain CO_2 assimilation, allowing other physiological or environmental factors to become the next limiting step in growth. Identifying this threshold will require further investigation and is likely to vary depending on crop species and cultivation system.

3.5 Conclusion

This study evaluated the effects of elevated humidity levels, lighting spectra, and air velocity on the morphological and physiological development of lettuce cultivated in a controlled environment. Average air velocity experienced by a given plant ranged from 0 to 3 m s^{-1} , with no significant correlations between growth or gas exchange observed. Tipburn severity and occurrences were minimized across all treatments by air movement, correlating with effective transpiration and stomatal function even at 90% RH. Although humidity alone did not significantly impact plant growth, these findings suggest that normal lettuce development can be maintained within 70–90% RH (VPD: 0.2–0.7 kPa). Strategic air movement may allow this range to be extended further. Amber wavelengths enhanced certain morphological traits like leaf area, but simultaneously reduced chlorophyll accumulation. These effects were generally more pronounced at 70% RH, but FR photons may steer plant development towards leaf expansion at elevated humidities. Far-red supplementation paired with air movement may have facilitated the greatest enhancements in assimilation under WFR despite its low R:B ratio. Overall, this study demonstrates the effectiveness of air movement at attenuating deleterious effects of elevated humidity through practical elimination of stomatal dysfunction as a limiting factor to plant growth and expansion. Unique plant responses to specific lighting spectra are dampened under varying air velocities, and further research will likely require lighting spectra with extremely pronounced spectral features to induce any distinct differences in plant growth. However, this study provides researchers and growers with a baseline to push plant growth even higher through further optimizations of environmental factors in controlled environments without fear of plant dysfunction.

Connecting Statement to Chapter 4

The previous chapter explored the effects of varying humidity levels, lighting spectra, and air velocity on the morphological and physiological characteristics of lettuce cultivated in a controlled environment. The following chapter will discuss this study's contributions to the academic body of knowledge, highlight key limitations that may influence the generalizability of the findings, and consider broader implications for the CEA industry.

Chapter 4 – General Discussion

The goal of this study was to evaluate the effects of heightened humidity on lettuce cultivation in controlled environments. Vertical farms and plant factories often face difficulties controlling humidity due to a combination of limited physical dimensions and high-density lettuce cultivation that can overwhelm cooling and dehumidification systems via net transpiration from plants (Graamans et al., 2017). Consequently, humidity can rise to unmanageably high levels, risking dysfunctions in plant physiological behavior and heightening susceptibility to pathogenic infections (Fanourakis et al., 2020; Su et al., 2004). Elevated humidity can be especially pronounced at the microclimate level near the plant canopy, where humidity and temperature are potentially higher compared to the ambient environment (Fanourakis et al., 2020). This study offered substantial insight into the role air velocity has on promoting normal lettuce growth even under suboptimal humidity levels. Amber wavelengths, FR, and low R:B ratio each induced unique morphological changes even under elevated humidity.

4.1 Contributions to Knowledge

Results of this study provided further information about the role air velocity plays in plant physiological function and its interactions with different lighting and elevated humidity treatments. Prior studies on leafy green production in controlled environments either focus primarily on the validation of a previously simulated air movement system for a given system or examining limited interactions between velocity and a single environmental factor. Comprehensive reporting of morphological and physiological growth parameters is limited, with most studies emphasizing reductions in tipburn severity and reduction as the primary result (Kaufmann, 2023; Peiro et al., 2020). Additionally, no studies of this nature have analyzed elevated humidity levels as an environmental factor despite the high possibility of humidity control malfunction in plant factory settings (Graamans et al., 2017). Therefore, the results of this study aim to provide future researchers with a baseline for plant function at suboptimal humidity levels.

Results further clarify the roles of amber and far-red wavelengths in plant development. While amber-biased high-pressure sodium (HPS) lamps are widely used in greenhouse cultivation, the specific effects of amber wavelengths remain underexplored (Wu et al., 2021). In particular, influence of amber wavelengths on gas exchange performance in plants remains poorly characterized. Observed parallels between amber, green, and far-red wavelengths suggest potential interaction effects that merit further investigation on leafy green cultivation. While these interactions are likely synergistic, they could introduce tradeoffs in plant function such as reductions in leaf thickness or water-use efficiency. Future research in controlled environments needs to include amber wavelengths to expand current understanding regarding this underexplored region of the PAR spectrum.

4.2 Limitations

Several promising avenues for future research were identified, including investigations into leaf thickness and water-use efficiency under wide amber and far-red wavelengths. Results also provided valuable context to previous studies of this nature, directly addressing potential shortcomings for air movement in CEA. However, several significant limitations must be acknowledged within the current study, as they may negatively affect interpretations of the findings.

4.2.1 Current Study

A potential flaw in experimental design within this study could have been the timing of gas exchange measurements, especially in regard to assimilation. Assimilation measurements were taken at the end of the 28-d growth cycle, meaning that plants were close to or at maturity and leaves selected would have been older leaves. At this stage, there is greater accumulation of non-structural carbohydrates such as starch and sucrose in the leaves. This buildup may have contributed to the repression of photosynthesis, a response that has been reported across various crop species (Jeannette et al., 2001; Krapp & Stitt, 1995). The effect of repression is greater in older, more mature leaves where decreases in Rubisco and chlorophyll content have been correlated with higher concentrations of soluble sugars (Araya et al., 2006).

Transpiration is less likely to be affected by this accumulation, as movement of moisture is governed through natural differences water vapor pressure between the leaf and ambient air as opposed to photosynthetic mechanisms (Spomer & Tibbitts, 1997). Similarly, stomata responses generally relate to changes in humidity or lighting (Fanourakis et al., 2020; Rufyikiri et al., 2024). While studies studying carbohydrate accumulation influence on photosynthesis have yet to be conducted on lettuce cultivated in controlled environments, this response could potentially account for the lack of significant differences in assimilation between different humidity and lighting treatments. Further study on temporal variations in gas exchange throughout the growth cycle in CEA is warranted.

Two previous studies by Adamarola (pending publication) and Samy, which employed similar growing techniques and methodologies, reported lettuce mass nearly double those observed in this study (Samy, 2023). Although each study had a distinct focus—Adamarola on CO₂ concentration and Samy on planting density—this disparity in fresh mass cannot be fully explained by treatment differences alone. The significant reduction in fresh mass is likely due to planting density. In this study, a 10.2 cm non-staggered configuration was used to acquire more plants for data collection. Samy reported that romaine lettuce grown under this configuration had an average fresh mass of 111.1g 71 d after seeding, while this study harvested 49 days after seeding with fresh mass ranging from 81-91 g (Samy, 2023). The densities that had the highest yields were 15.2 cm and 17.7 cm non-staggered, similar to values utilized by Adamarola. Therefore, higher density combined with a lower growth time to harvest are likely to have reduced average fresh yield per plant, emphasizing the negative influence of high planting density on lettuce development. While the specific effects of density in this study cannot be isolated due to major effects like lighting and humidity, results could potentially open pathways for alternative growing strategies within CEA facilities.

4.2.2 Prior Studies

Previous studies evaluating the effects of air movement on plant growth in plant factories or vertical farms often correlate increases in air velocity to improvements in morphology, biomass accumulations, and gas exchange performance. While these studies

consistently report reductions in tipburn occurrences and severity, the methodologies and reporting relating to air velocity distribution and gas exchange measurements are often insufficient or inconsistently detailed. Consequently, conclusions suggesting that increased air velocity directly promote biomass accumulation and enhanced assimilation may need to be reexamined.

Air velocity distribution is normally reported as a steady-state value at a standard distance relative to the cultivation tray regardless of the air movement system (Kaufmann, 2023; Kitaya et al., 2004; Peiro et al., 2020). This trend is likely due to the utilization of CFD models to evaluate air velocity distribution of a given air movement system in vertical farms. CFD is a powerful tool that allow growers to account for all potential factors that could affect air flow in their system, and results of these models are often in good agreeance with validation tests (Sohn et al., 2023; Zhang et al., 2016). Simulations results are generally reported as steady state values, meaning the air velocity that a plant undergoes will inevitably shift in both magnitude and direction as it grows and develops.

Reporting of the variations in actual air velocity experienced by a plant over time often goes neglected. This is concerning, particularly if mature plants are exposed to excessively high velocities that may hinder rather than promote growth. The current study did not observe any reductions or improvements in plant performance at air velocities up to 3.0 m s^{-1} . However, reported values in this study represented the average air velocity experienced by the plants over their 28-day growth cycle. As such, prior studies that report reductions in performance past certain thresholds may actually be significantly underreporting the actual air velocities plants were exposed to. These values may have been high enough such that plants begin to exhibit thigmomorphogenic defense mechanisms, therefore depressing plant growth. Further studies that apply air movement need to address how air velocity shifts over time at canopy level to ensure accurate characterization of the microclimate in the cultivation tray experienced by the plant, providing further context to morphological and physiological development.

Air movement is well established to reduce tipburn severity and occurrences, but correlations to transpiration rates are shaky (Ahmed et al., 2020b). Reporting of tipburn reduction is often used in lieu of transpiration rates, likely due to technological limitations that

make direct quantification of moisture movement difficult without the use of a portable gas exchange measurement system or mathematical transpiration models like the Penman-Monteith model. While the Penman-Monteith model has been frequently used in biometeorological studies to evaluate net transpiration of large tracts of land, its effectiveness is reduced when applied on an individual plant basis (Wehr & Saleska, 2021). This is due to its simplifying assumption that consolidates an entire canopy as a single “big leaf”, which consolidates variation in microclimates between individual plants (Wehr & Saleska, 2021). Models have been proposed that integrate the Penman-Monteith equation as part of a full energy balance in greenhouses, but these models have not been applied for direct transpiration measurement in any studies, let alone for indoor factories or vertical farms (Ahamed et al., 2018; Graamans et al., 2017; Pollet et al., 2000). Stomatal conductance is an even more difficult parameter to estimate without direct measurement, and has to be derived indirectly using the latent-heat flux term in the Penman-Monteith equation that is prone to biases (Wehr & Saleska, 2021).

Net CO₂ assimilation can be derived using inlet and outlet CO₂ concentrations combined with volumetric air exchange rate of a chamber as described by Kitaya (Kitaya et al., 2004). This methodology suffers similar weaknesses to the “big leaf” model, masking individual plant responses in a growth tray and making it impossible to identify microclimate influence on net assimilation.

Individual plant photosynthetic response to varying environmental factors is an extensive area of research, and photosynthetic mechanisms are still being explored (Hogewoning et al., 2012; Wu et al., 2025). Future studies in controlled environment facilities need to consider physiological performance of plants on an individual basis to accurately make determinations about the overall performance of a particular system. Models such as the Penman-Monteith equation are useful for approximations when vital crop information is missing, but a more direct approach is required to avoid the weaknesses inherent to these models.

4.3 Implications for Industry

The CEA industry in recent years has quickly become a high-profile industry, attracting large amounts of investors and resources across North America and the globe. However, volatility in the industry is still the norm, as high initial capital and high energy demand from LED lighting and HVAC systems often prove too large a burden on smaller vertical farms (Benke & Tomkins, 2017; Walling & Lafleur, 2023). Expansion of CEA worldwide will continue to be a difficult economic dilemma to resolve due to reliance on low energy costs (Cowan et al., 2022). Therefore, growers will need to adjust their growing strategies based on a variety of market forces.

The results of this study emphasize an important point for growers: Air movement allows lettuce growth to not only be preserved, but continually optimized even under elevated humidity levels. HVAC systems and dehumidification are some of the most energy-intensive factors in operating vertical farms along with artificial lighting (Cowan et al., 2022). Control of temperature and humidity is even more difficult due to net energy input from plants overwhelming HVAC systems, causing environmental control to suffer (Graamans et al., 2017). By implementing an air movement system, lettuce growth can be maintained without suffering physiological disorders like tipburn even under elevated humidity. Growing strategies can then be tailored to economic goals as opposed to worrying about the impacts on plant health. For example, some growers may opt for lower planting densities to allow individual plants more space to grow to larger sizes. While edge effects can hinder growth due to competition for resources within a tray, this study suggests that high planting densities can still produce plants of comparable quality. Further optimization of lighting spectra can be tested to adjust morphological traits to desired quality. Regardless, air movement systems would eliminate plant transpiration or stomatal resistances as limiting factors to growth, offering growers the flexibility to implement other environmental optimizations aimed at promoting plant uniformity, quality, and yield.

Chapter 5 – Overall Conclusion

This thesis evaluated the effects of elevated humidity levels, lighting spectra, and air velocity on the morphological and physiological development of lettuce cultivated in a controlled environment. Average air velocity experienced by a given plant ranged from 0.23 to 3 m s⁻¹, with no significant correlations between growth or gas exchange observed ($p > 0.05$). However, tipburn severity and occurrences were effectively minimized across all treatments even under relative humidity levels as high as 90%. Therefore, an air movement system that blows air from 0.5–1 m s⁻¹ at tray level can be used to maintain normal plant function and growth within 70–90% RH (VPD: 0.2–0.7 kPa), with potential to extend this range further. While lighting treatments did not have strong effects on plant development, certain morphological differences indicate the potential of both amber and far-red wavelengths. Amber wavelengths enhanced certain morphological traits like leaf area, where WA at 70% RH ($123.7 \pm 16.77 \text{ cm}^2$) and was significantly higher than WFR at 80% RH ($97.0 \pm 13.62 \text{ cm}^2$; $p < 0.05$) and 90% RH ($99.58 \pm 9.25 \text{ cm}^2$; $p < 0.05$). However, WA had significantly lower chlorophyll SPAD count at 70% RH, although total chlorophyll accumulation was not significantly different between lighting treatments ($p > 0.05$). These effects were generally more pronounced at 70% RH, but far-red photons may steer plant development towards leaf expansion at elevated humidities. Far-red supplementation paired with air movement may have also facilitated the greatest enhancements in assimilation under WFR, leading to no significant differences in gas exchange between any lighting treatment ($p < 0.05$) despite a much lower R:B ratio under WFR (0.15) compared to WA (1.56) and WL (0.62).

Overall, this thesis demonstrated the effectiveness of air movement at attenuating deleterious effects of elevated humidity through practical elimination of stomatal dysfunction as a limiting factor to plant growth and expansion. Unique plant responses to specific lighting spectra were dampened under varying air velocities, but air movement leaves substantial room for further optimizations of other environmental factors. Results presented offer a more comprehensive understanding of plant growth under air movement in controlled environments than previous studies of this nature. This study established a valuable baseline for future research, allowing for the introduction of more pronounced treatments to better observe plant

response. Furthermore, these insights are vital for helping growers maintain lettuce quality and yield in CEA facilities such as plant factories or vertical farms, which are particularly susceptible to humidity control issues due to high-density cultivation and improperly sized cooling or dehumidification systems.

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Appendix A – Germination Schedule

Table A1. Electrical conductivity (EC) and pH monitoring during germination (21 d)

Days	EC ($\mu\text{S cm}^{-1}$)	pH
0–5	310	5.0
6–9	423	5.0
10–12	510	5.1
13–15	623	5.2
16–17	717	5.2
18–21	900	5.1

Appendix B – Growth Room Performance

Figure B1 shows the performance of each environmental parameter in each growth chamber over a 30-d period for Trial 5 (T5) as an example. Large spikes in any variable are generally due to researchers entering the room to take samples/measurements; opening doors causes humidity to drop as moisture escapes, while researchers staying inside causes CO₂ to spike due to their exhalation(s). Table B1 shows the average and standard deviation for each variable for the same trial. While performance in each chamber varied due to how each room was constructed and/or used, overall, environmental parameters were kept at very stable levels over each trial.

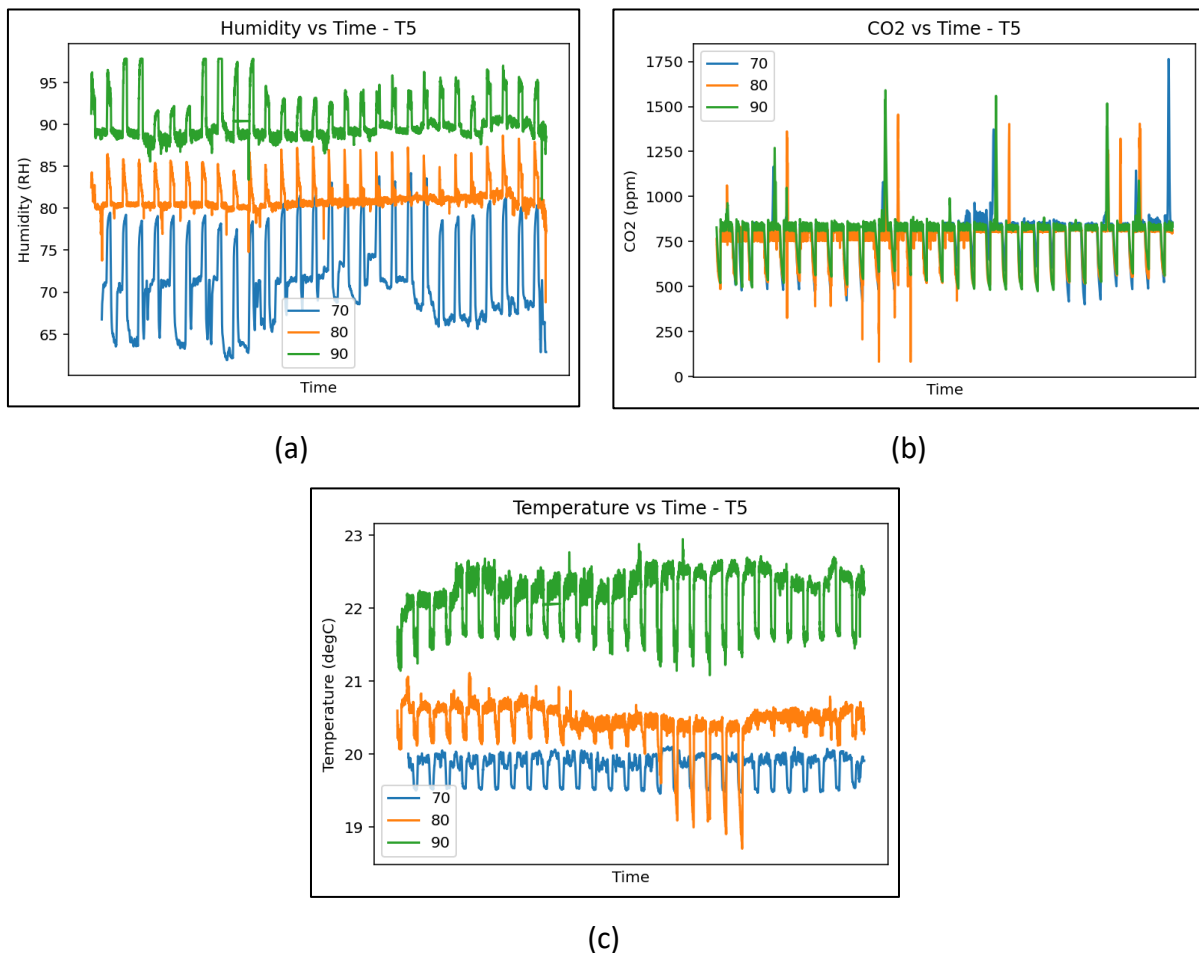


Figure B1. Environmental control performance in each growth chamber for (a) humidity, (b) CO₂, and (c) temperature

Table B1. Average and standard deviation for each variable

Chamber RH	Temperature	Humidity	CO ₂
70%	19.83 ± 0.20	71.31 ± 5.36	799.33 ± 134.44
80%	20.43 ± 0.28	81.28 ± 1.57	754.44 ± 121.28
90%	22.17 ± 0.35	90.26 ± 2.35	786.87 ± 119.58

Appendix C – Regression and Model Analyses

Table C1. Regression and model analyses for growth parameters. p-values represent the best fitting model where non-significant terms were removed ($p > 0.05$). Rows ordered by adjusted R^2 .

Growth Parameter	Effect Test (Significance)				Model Analysis		
	Lighting	Humidity	Lighting*Humidity	Humidity*Humidity	ANOVA	R^2	Adjusted R^2
Chlorophyll a	0.97	0.23	0.80	0.87	0.90	0.09	-0.18
Carotenoids	0.56	0.38	0.79	0.97	0.86	0.10	-0.16
Chl a:b	0.82	0.34	0.79	0.39	0.85	0.11	-0.15
Chlorophyll b	0.90	0.21	0.90	0.36	0.80	0.13	-0.13
Specific Leaf Area ($\text{cm}^2 \text{g}^{-1} \text{dry mass}$)	0.73	0.19	0.72	0.65	0.79	0.13	-0.12
Anthocyanins	0.60	0.93	0.34	0.47	0.70	0.15	-0.09
Dry Root Mass (g)	0.66	0.18	0.21	0.95	0.44	0.23	0.00
Fresh Root Mass (g)	0.39	0.06	0.90	0.23	0.06	0.13	0.09
Dry Mass (g)	0.25	0.38	0.92	0.41	0.61	0.18	0.05
Root Mass Ratio (RMR)	0.19	0.09	0.32	0.67	0.21	0.31	0.11
Fresh Mass (g)	0.07	0.43	0.83	0.10	0.19	0.32	0.12
Head Diameter (cm)	0.03	0.09	0.27	0.85	0.03	0.23	0.17
Leaf Area (cm^2)	0.01	0.61	0.26	0.25	0.01	0.28	0.22
Number of Leaves	0.15	< 0.01	0.75	0.64	< 0.01	0.28	0.25
Chlorophyll (SPAD)	< 0.01	0.02	0.96	0.86	< 0.01	0.47	0.40
Height (cm)	0.04	< 0.01	0.28	0.82	< 0.01	0.53	0.47

Table C2. Regression and model analyses for gas exchange parameters. p-values represent the best fitting model where non-significant terms were removed ($p > 0.05$). Rows ordered by adjusted R^2 .

Growth Parameter	Effect Test (Significance)				Model Analysis		
	Lighting	Humidity	Lighting*Humidity	Humidity*Humidity	ANOVA	R^2	Adjusted R^2
Transpiration ($\text{mmol m}^{-2} \text{s}^{-1}$)	0.63	0.66	0.89	0.22	0.81	0.12	-0.13
Assimilation ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	0.56	0.25	0.81	0.68	0.78	0.13	-0.12
Stomatal Conductance ($\text{mol m}^{-2} \text{s}^{-1}$)	0.69	0.61	0.91	0.14	0.73	0.14	-0.1

Appendix D – Air Velocity Mapping Details

Air velocity mapping was conducted to determine how the air velocity experienced at plant level changed throughout the growth cycle, as air velocity distributions are normally reported only at the start of the growth trial. Air velocity distribution was measured four times per trial (M0 – M3), after which the average velocity, standard deviation, and coefficient of variation were calculated per each combination of humidity, lighting, and velocity. Table D1 and Figure D1 present the overall average air velocity for each plot across all replicates, while Table D2 and Figure D2 show the change in average air velocity by velocity zone week by week.

Table D1. Overall air velocity statistics across all replicates

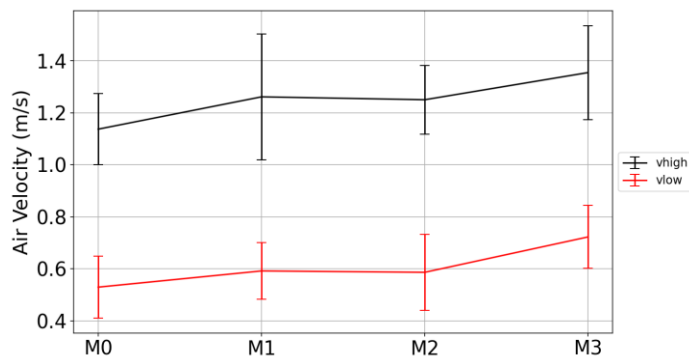
Velocity		70%		80%		90%	
		Average	CV	Average	CV	Average	CV
WA	High	1.31 ± 0.27	0.20	1.48 ± 0.36	0.24	1.30 ± 0.37	0.29
	Low	0.63 ± 0.07	0.12	0.74 ± 0.16	0.22	0.60 ± 0.16	0.27
WFR	High	1.13 ± 0.13	0.12	1.32 ± 0.39	0.29	1.53 ± 0.47	0.31
	Low	0.57 ± 0.05	0.09	0.65 ± 0.15	0.23	0.68 ± 0.15	0.22
WL	High	1.34 ± 0.18	0.13	1.50 ± 0.23	0.16	1.38 ± 0.35	0.25
	Low	0.62 ± 0.15	0.24	0.76 ± 0.19	0.25	0.63 ± 0.06	0.09

Lighting	Velocity	70%			80%			90%		
WA	High	1.28		1.20	1.21		1.31	0.96		1.00
		1.38	1.67	1.14	1.38	2.07	1.33	1.08	1.91	1.23
		1.42	1.38	0.88	1.55	1.58	0.97	0.99	1.57	0.99
	Transition	0.71	0.79	0.63	0.83	1.01	0.62	0.62	0.71	0.64
		0.72	0.67	0.60	0.89	0.94	0.70	0.60	0.90	0.63
		0.52	0.68	0.59	0.57	0.78	0.54	0.49	0.57	0.43
	Low	0.32		0.34	0.33		0.33	0.32		0.34
WFR	High	1.09		1.26	1.40		1.42	1.27		1.31
		1.20	1.17	1.17	1.61	1.91	1.20	1.22	2.17	1.46
		1.30	1.01	0.94	1.03	1.31	0.84	1.03	2.06	1.24
	Transition	0.66	0.68	0.65	0.78	0.96	0.70	0.70	0.93	0.91
		0.58	0.66	0.53	0.60	0.84	0.65	0.69	0.73	0.89
		0.56	0.51	0.58	0.43	0.80	0.57	0.54	0.76	0.49
	Low	0.34		0.37	0.31		0.27	0.34		0.34
RV	High	1.28		1.13	1.33		1.10	0.95		1.02
		1.28	1.65	1.13	1.54	1.71	1.29	0.99	1.81	1.20
		1.39	1.34	1.22	1.48	1.79	1.19	1.04	1.67	1.56
	Transition	0.80	0.71	0.62	0.94	0.87	0.69	0.53	0.71	0.68
		0.82	0.75	0.58	0.73	0.82	0.64	0.56	0.68	0.70
		0.53	0.57	0.43	0.58	1.12	0.69	0.58	0.67	0.61
	Low	0.30		0.25	0.39		0.35	0.46		0.64

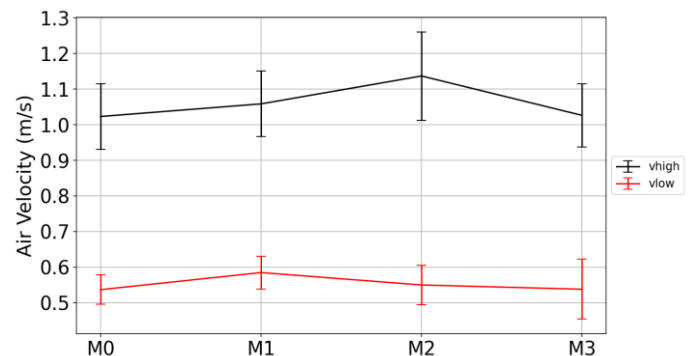
Figure D1. Average overall air velocity across all replicates

Table D2. Change in air velocity over time across all replicates

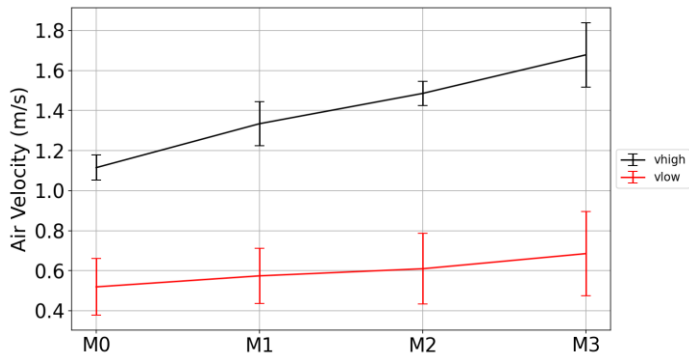
Humidity	Lighting	Velocity	M0	M1	M2	M3	M0-M3 (%)
70	WA	Low	0.53 ± 0.12	0.59 ± 0.11	0.59 ± 0.15	0.72 ± 0.12	36.48
		High	1.14 ± 0.14	1.26 ± 0.24	1.25 ± 0.13	1.35 ± 0.18	19.14
	WFR	Low	0.54 ± 0.04	0.58 ± 0.05	0.55 ± 0.06	0.54 ± 0.08	0.23
		High	1.02 ± 0.09	1.06 ± 0.09	1.14 ± 0.12	1.03 ± 0.09	0.31
	WL	Low	0.52 ± 0.14	0.57 ± 0.14	0.61 ± 0.18	0.69 ± 0.21	31.96
		High	1.12 ± 0.06	1.33 ± 0.11	1.49 ± 0.06	1.68 ± 0.16	50.54
80	WA	Low	0.63 ± 0.22	0.65 ± 0.11	0.74 ± 0.14	0.88 ± 0.19	39.31
		High	1.25 ± 0.15	1.36 ± 0.23	1.54 ± 0.25	1.78 ± 0.36	42.30
	WFR	Low	0.57 ± 0.09	0.68 ± 0.10	0.60 ± 0.08	0.74 ± 0.17	31.03
		High	1.10 ± 0.32	1.32 ± 0.46	1.27 ± 0.36	1.67 ± 0.29	51.74
	WL	Low	0.58 ± 0.09	0.66 ± 0.07	0.78 ± 0.08	0.92 ± 0.08	57.64
		High	1.24 ± 0.11	1.29 ± 0.09	1.41 ± 0.18	1.54 ± 0.25	24.50
90	WA	Low	0.48 ± 0.04	0.54 ± 0.11	0.55 ± 0.15	0.78 ± 0.32	61.51
		High	1.00 ± 0.16	1.09 ± 0.23	1.14 ± 0.09	1.81 ± 0.20	80.69
	WFR	Low	0.54 ± 0.11	0.64 ± 0.22	0.67 ± 0.22	0.87 ± 0.10	60.99
		High	1.20 ± 0.05	1.46 ± 0.09	1.52 ± 0.11	1.90 ± 0.21	59.18
	WL	Low	0.48 ± 0.04	0.58 ± 0.07	0.62 ± 0.08	0.68 ± 0.05	40.61
		High	1.08 ± 0.17	1.31 ± 0.13	1.34 ± 0.09	1.75 ± 0.11	62.00



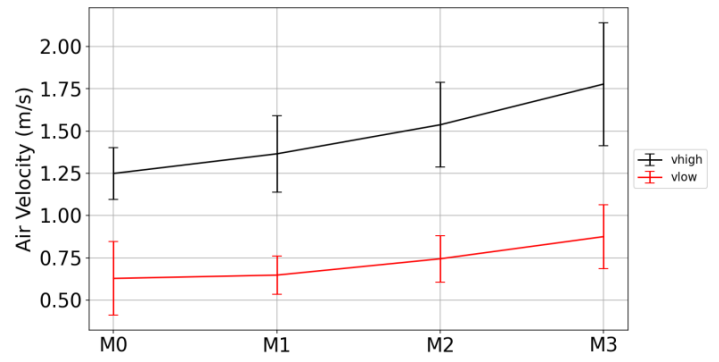
(a)



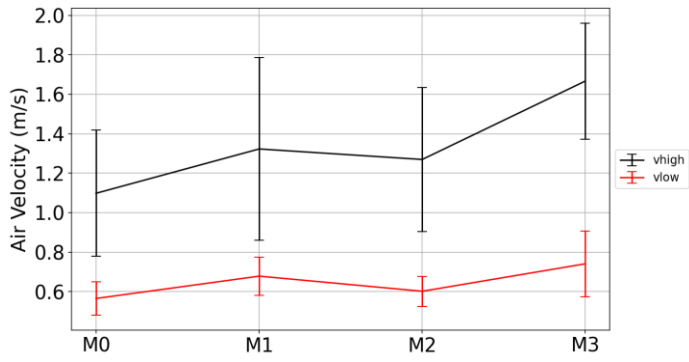
(b)



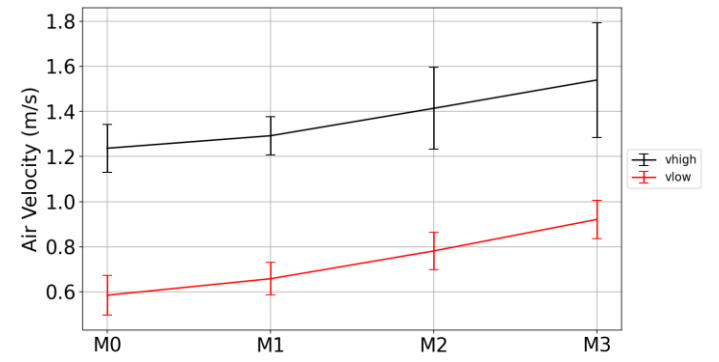
(c)



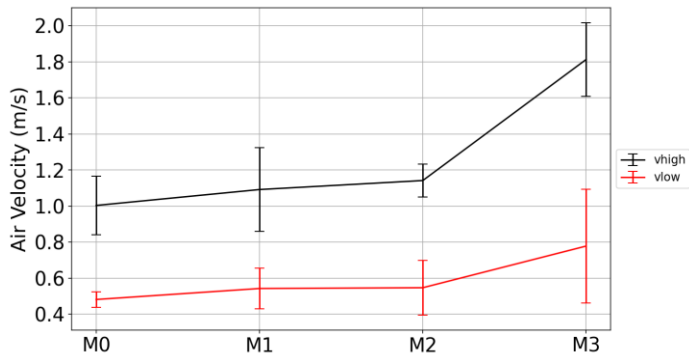
(d)



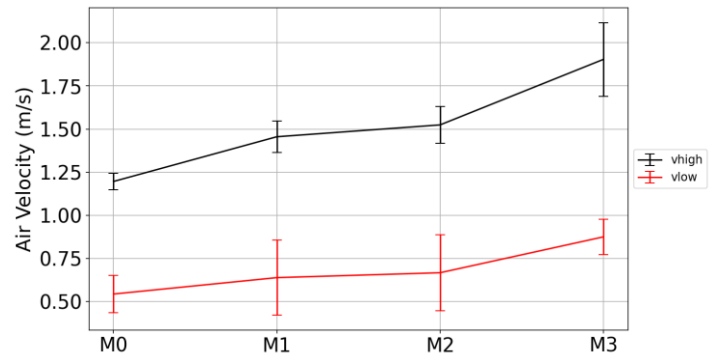
(e)



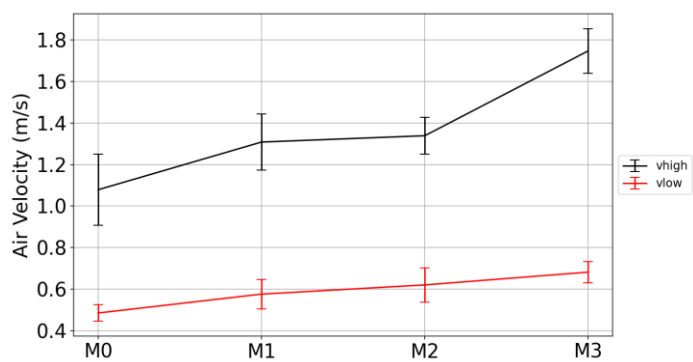
(f)



(g)



(h)



(i)

Figure D2. Plotted changes in air velocity over time across all replicates. 70% relative humidity (RH) and (a) Wide Amber (WA) (b) White + FR (WFR) and (c) White Light (WL). 80% RH for (d) WA (e) WFR and (f) WL. 90% RH for (g) WA (h) WFR and (i) WL.

Appendix E – Air Velocity Regression

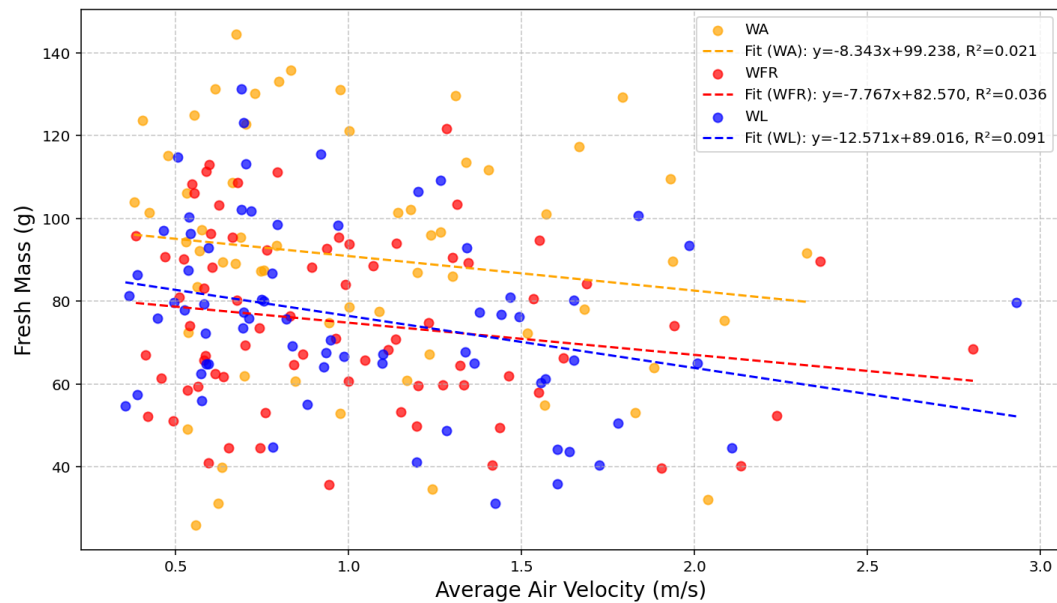


Figure E1. Fresh mass vs average air velocity across all replicates grouped by lighting. [WA: wide amber supplemented with far-red, WFR: white light supplemented with far-red, WL: white light. RH: relative humidity]

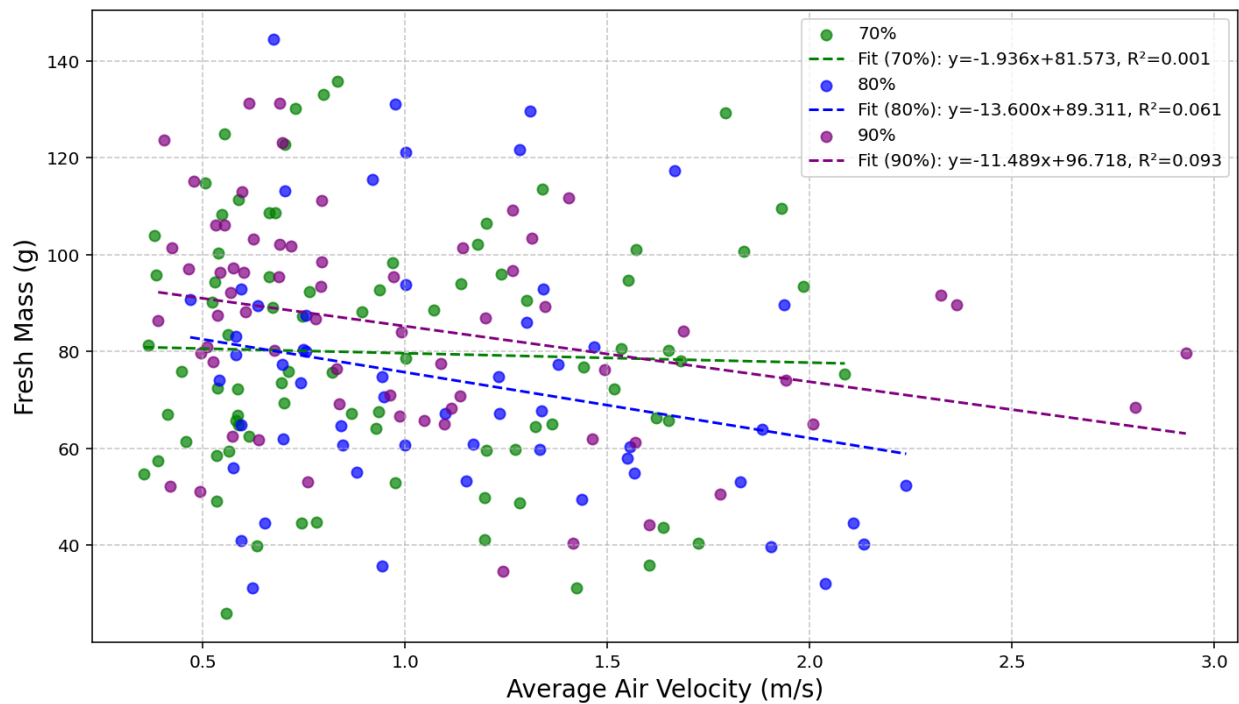


Figure E2. Fresh mass vs average air velocity across all replicates grouped by humidity